SICHUAN UNIVERSITY

Undergraduate Graduation Project
(Academic Thesis)

Title	**Transformer-Based Approaches for Automated Detection of Suicidal Ideation on Social Media**
School	**College of Software Engineering**
Major	**Software Engineering**
Student's Name	**Shakib Al Hasan**
Student ID:	**2021521460124** **Grade 2021**
Adviser	**Dasha Hu**

Made by the Office of Academic Affairs

May 20, 2025

ABSTRACT

Software Engineering

Student: Shakib Al Hasan Adviser: Dasha Hu

[Abstract] Suicide remains a priority global health issue, claiming over 700,000 lives annually and ranking as the fourth leading cause of death among 15–29 year-olds. This study addresses a critical research gap by conducting the first comprehensive cross-paradigm evaluation of suicide detection systems on social media. We compare ten computational models across three algorithmic approaches (traditional machine learning, deep learning, and transformer architectures) using a unified dataset of 232,074 Reddit comments. Our novel contributions include: (1) a specialized preprocessing pipeline that preserves psychological distress markers while addressing class imbalance, (2) a rigorous comparative framework employing confusion matrices and ROC curves alongside conventional metrics, and (3) empirical evidence demonstrating transformer models' superior performance. BERT significantly outperformed all alternatives, achieving 97.62% accuracy and 97.63% F1-score, demonstrating bidirectional encoding's effectiveness in capturing nuanced linguistic patterns associated with suicidal ideation. These findings establish transformer models as a promising foundation for developing ethically sound, high-sensitivity suicide prevention systems that balance detection capabilities with privacy protection and human oversight.

[Key Words] suicide detection, transformer models, BERT, deep learning, social media analysis, mental health monitoring, natural language processing, early intervention

ABSTRACT

Software Engineering

Student: Shakib Al Hasan Adviser: Dasha Hu

摘要

自杀仍然是全球健康问题中的优先事项，每年造成超过70万人死亡，并且在15至29岁人群中排名第四大死亡原因。本研究通过首次对社交媒体上的自杀检测系统进行跨范式的全面评估，填补了一个关键的研究空白。我们比较了十种计算模型，涵盖了三种算法方法（传统机器学习、深度学习和变换器架构），并使用了一个统一的数据集，包括232,074条Reddit评论。我们的创新贡献包括：（1）一种专门的预处理流程，既能保留心理困扰的标记，又能解决类别不平衡问题；（2）采用混淆矩阵和ROC曲线以及常规指标的严格比较框架；（3）实证证据表明，变换器模型表现优异。BERT显著优于所有其他方法，达到了97.62%的准确率和97.63%的F1分数，展示了双向编码在捕捉与自杀意图相关的细微语言模式中的有效性。这些发现确立了变换器模型作为开发伦理可靠、高敏感度的自杀预防系统的有希望的基础，这些系统能够在检测能力、隐私保护和人工监督之间取得平衡。

关键词：自杀检测，变换器模型，BERT，深度学习，社交媒体分析，心理健康监测，自然语言处理，早期干预

Table of Contents

1 Preamble

1.1 Introduction

One of the most urgent worldwide public health issues of our time, suicide The World Health Organization estimates that more than 800,000 people die by suicide each year, or one death every 40 seconds. Suicide is the fourth most common cause of death among people between the ages of 15 and 29 worldwide [1]. Suicide rates are rising in many nations, including a 35% rise in the United States between 1999 and 2018 [2] even with more prevention initiatives under this crisis. On the other hand, some countries like China have seen extraordinary success in lowering suicide rates, with research showing a notable drop from about 23.0 per 100,000 population in 1999 to 8.6 per 100,000 population in 2017, representing a 63% reduction over the last two decades [3].

Social media's broad utilize has changed the way individuals communicate mental trouble and created computerized follows that show already unheard-of chances for early mediation. Thinks about uncover that weeks or months earlier to suicide endeavors, social media substance regularly contains phonetic pointers of self-destructive ideation [4]. These digital expressions are important sources of data that may be able to identify people who are at risk during crucial times when intervention may be most successful.

While pharmacological and psychotherapy treatments are still essential components of suicide prevention strategies, their reaction nature significantly limits their prevention potential [5]. The integration of advanced calculation technologies with health care initiatives offers transforming opportunities to prevent active suicide. Digital technologies allow analysis to systematize a large amount of patient and population data, capable of identifying high -ratio people before reaching crisis points [6].

The recent process of artificial intelligence, especially in learning techniques, has proven noticeable ability in analyzing non -structural data to monitor mental health. Natural language treatment algorithms can detect depression signs and suicide ideas in social networking publications with increasing reliability [7]. The usual automatic learning method using the supporting vector machine (SVM), random forest and logistics regression has succeeded in moderate success in this field, but limited by their dependence on manual modifications and cannot grasp the semantic context effectively [8].

With improved context understanding and semantic processing skills, transformer-based models provide considerable breakthroughs in understanding natural language. These designs can identify unconscious indicators that overlook traditional analyses due to their small linguistic elements that may indicate suicide intentions and their extraordinary ability to identify emotional information [9]. Bidirectional models such as BERT (Bi-directional encoder representation of transformers) simulate a complete understanding of human linguistic processing by evaluating each word in the context of both words that come forward and backward.

By contrasting state-of-the-art transformer models with conventional machine learning and deep learning models, this thesis assesses the effectiveness of various architectures for constructing an intelligent system for detecting suicidal tendencies. By analyzing text information from social networking sites, our goal is to craft an effective framework capable of precisely pinpointing vulnerable individuals while minimizing incorrect alerts. Employing transfer learning techniques to optimize functionality across diverse environments, the proposed system has the potential to serve as a scalable early intervention resource for preventing suicide.

This research is rising to advances in its technological advances. Existing prevention methods can be significantly improved, and individuals at risk can receive timely support by determining reliable automated mechanisms for recognizing suicide thoughts and behaviors. Technological approaches provide an alternative method of communicating with individuals who may go unnoticed until they are experiencing a crisis, particularly in regions where mental health resources are scarce and social pressures prevent many individuals from receiving the necessary assistance.

1.2 State of the Art

1.2.1 Theoretical Foundations in Computational Suicide Detection

Existing psychological theories that simulate suicidal behavior serve as the foundation for computational methods for suicide detection. The primary predictors of suicidal ideation, according to Joiner's Interpersonal Theory of Suicide, are perceived burdensomeness and thwarted belongingness [10]. O'Connor's Integrated Motivational-Volitional Model provides a comprehensive framework for understanding the transition from suicidal ideation to suicidal behaviors, emphasizing the role of entrapment and defeat in this process [11]. These theoretical frameworks have served as a guide for the selection of features and the development of models in computational approaches to suicide detection.

Pestian et al. established the foundation for the identification of linguistic indicators associated with suicidal ideation by demonstrating distinct lexical and semantic patterns in suicide notes that differentiate authentic notes from fake ones [12]. In subsequent research, Coppersmith et al. [13] extended these findings to social media contexts, observing similar linguistic patterns in online content that preceded suicide attempts. Some of these patterns include the increasing use of first-person pronouns, absolutist reasoning, and dejection; these indicators are now essential to computational suicide detection systems.

1.2.2 Classical Machine Learning Approaches

The primary foundation of early computational approaches to suicide detection was conventional machine learning algorithms that were meticulously designed with features. Burnap et al. were the first to apply Support Vector Machines and Random Forests to the classification of Twitter posts that convey suicidal ideation developed methodological frameworks that influenced subsequent research [14]. O'Dea et al. enhanced these methods by employing Logistic Regression models to distinguish between genuine suicidal intent and general distress [15].

These conventional methodologies typically rely on feature sets that have been meticulously developed. These feature sets consist of linguistic identifiers (sentiment, tense, and pronoun usage), psycholinguistic features (emotional tone, cognitive processes), and behavioral indicators (posting frequency, temporal patterns). Despite the initial promise of these techniques, they are inherently incapable of capturing the linguistic subtleties and contextual complexity that are characteristic of suicidal expression [16].

1.2.3 Deep Learning Architectures

Implementation of deep learning architectures has shown significant advances in the identification of suicide. Repeated neural networks (RNNs) and long-term short-term memory networks (LSTMs) represent early innovations in this domain, allowing them to process temporary data in text and capture temporal dependencies [17]. These structures exhibited a greater capacity to replicate the contextual connections that are essential for comprehending textual emotional expression.

Convolutional Neural Networks (CNNs) have also made valuable contributions to the field of suicide detection research, particularly in their ability to identify local patterns in text data that suggest suicidal intent [18]. These models are particularly effective in identifying specific words and linguistic patterns that are associated with suicide ideation. However, they may fail to consider more general background signals that are necessary for a comprehensive evaluation.

1.2.4 Transformer-Based Models

The introduction of transformer-based architectures represents a paradigm shift in skills for processing natural language for recognition of self-disturbance. The self-combat mechanisms embedded in these models allow for the most wonderful context-related understanding and semantic processing, as well as important skills for recognizing the often foreign languages associated with suicidal thinking [19].

The nuanced linguistic markers of psychological distress have been captured with extraordinary effectiveness by BERT (Bidirectional Encoder Representations from Transformers) and its variants [20]. The bidirectional nature of these models enables them to consider both the preceding and subsequent context when interpreting each word, thereby emulating the comprehensive understanding that is characteristic of human language comprehension.

Recent research has focused on the development of transformer models that are specifically tailored for mental health applications. These domain-specific architectures capture emotional expressions and specific language patterns that may evade more generalized models [21]. For instance, hybrid architectures such as RoBERTa that are combined with LSTM or self-attention layers have demonstrated promising results in the identification of both local and global patterns in suicide-related text [20, 21]. The performance of these models has been further enhanced by the application of transfer learning methods, which enable the application of knowledge acquired in general language comprehension activities to the specific challenges of suicide detection.

1.2.5 Multimodal Approaches and Cross-Platform Analysis

Although text analysis continues to be the primary focus of computational suicide detection, recent research has investigated multimodal approaches that incorporate additional data streams. Research has shown that the detection capabilities are improved by the integration of linguistic features with behavioral indicators, such as temporal posting patterns and social interaction metrics [22]. These multimodal strategies provide a more comprehensive assessment system that allows you to recognize psychological distress that may have been overlooked in text analysis alone.

Another notable advance in this domain is cross-platform research. Gaur et al. We conducted a study that showed that combining data from several social media platforms achieves a more comprehensive understanding of psychological conditions rather than conducting a single platform survey [23]. This method recognizes different digital IDs that maintain individuals across the platform, as well as different modes of expression that are distinctive to different social media.

1.2.6 Ethical Considerations and Implementation Challenges

Development of computer development systems requires thorough ethical research and solving practical implementation challenges. In this area, open ethical concerns include data protection matters, declarations of consent, and possible bias in algorithms [24]. The study highlighted the need for transparent methods, user autonomy and fair intervention guidelines in the development of these systems.

Outcomes of false positives and false negatives continue to be constant problems in systems for suicide detection. Research has underlined that computer methods should enhance rather than supplant human evaluation, acting as decision support tools inside more comprehensive intervention systems [25]. Effective implementation relies on integration of automated detection systems with appropriate clinical response protocols. This is an important topic.

For these fundamentals, our study uses the capabilities of transformer-based models to create improved techniques for identifying suicidal tendencies on social media. Our work will address the shortcomings of current strategies and expand the changing scenes of computer methods for suicide prevention by including new architectural components.

1.3 Main Work of This Project

This project conducts a systematic comparative analysis of suicidal ideation detection in textual content, with particular focus on evaluating transformer-based models against traditional approaches. We leverage the bidirectional encoding capabilities of BERT and related architectures to identify subtle linguistic patterns associated with suicidal ideation. By analyzing social media text data, our research aims to enhance early detection methods for identifying at-risk individuals exhibiting signs of psychological distress.

The key methodological contributions of this research include:

- **Advanced Text Preprocessing Pipeline:** Development of a specialized preprocessing framework for mental health text data that preserves critical psychological distress markers while addressing the inherent class imbalance problem. Our approach incorporates sophisticated text cleaning techniques, contextual tokenization, and strategic dataset balancing that significantly enhances model performance across all architectural paradigms tested.

- **Multi-paradigm Comparative Framework:** Design and implementation of a comprehensive evaluation methodology that systematically assesses ten computational models across three distinct algorithmic approaches:

  - Traditional machine learning (Logistic Regression, SVM)
  - Deep learning architectures (LSTM, CNN-LSTM)
  - Transformer-based models (BERT, RoBERTa hybrids)

  This cross-paradigm analysis provides unprecedented insights into the relative strengths and limitations of each approach for the specific challenge of suicide ideation detection.

- **Precision-Recall Trade-off Analysis:** Implementation of a nuanced evaluation framework employing confusion matrices and ROC curves to assess model performance beyond simple accuracy metrics. This approach reveals critical insights into the sensitivity-specificity balance essential for responsible deployment in mental health applications, where both false positives and false negatives carry significant ethical implications.

Our experimental results demonstrate the exceptional performance of transformer-based models, with BERT achieving superior classification metrics (accuracy: 97.62%, precision: 97.28%, recall: 97.99%, F1-score: 97.63%) compared to traditional and deep learning approaches. The findings underscore how BERT's bidirectional encoding capability effectively captures contextual relationships and subtle linguistic patterns that indicate psychological distress—patterns that conventional models frequently miss. This performance advantage is particularly significant for early intervention systems, where the ability to recognize implicit expressions of suicidal ideation could substantially enhance mental health professionals' capabilities in identifying and supporting at-risk individuals.

1.4 Organization and Structure

This thesis is structured into six cohesive chapters that systematically present the development of our intelligent system for detecting suicidal tendencies using transformer models:

To facilitate readers' understanding of our system and enable secondary development, comprehensive technical documentation is provided in the appendices, including detailed requirement specifications, design documents outlining architectural decisions, software test documentation with evaluation protocols, user manuals for practical system operation, and well-documented source code for implementation.

Table 1.1. Thesis Structure and Chapter Descriptions

Ch.	Title	Description
I	Introduction	Establishes the research context by examining suicide as a global health issue (800,000+ deaths annually), explores the potential of social media analysis and transformer models for early detection, and outlines our comprehensive model comparison approach.
II	Background Knowledge	Presents a detailed review of suicidal ideation patterns on social media platforms, traditional ML approaches (SVM, Random Forest), deep learning architectures (CNN, LSTM), and transformer-based models (BERT, RoBERTa), with emphasis on their contextual understanding capabilities.
III	Methodology	Details our data collection approach using 232,074 Reddit text samples, preprocessing techniques including tokenization and lemmatization, and our balanced dataset creation with a 70-15-15% train-validation-test split strategy.
IV	System Implementation	Describes the implementation environment (Python, PyTorch, Hugging Face), technical architecture of the BERT-based system through flowcharts, and core modules: Input Processing, Embedding Layer, Transformer Encoder, and Output Layer.
V	Testing & Evaluation	Presents a rigorous comparative analysis of eleven models across three paradigms, demonstrating BERT's superior performance (97.62% accuracy, 97.63% F1-score) through confusion matrices and ROC curves, with detailed discussion of precision-recall trade-offs.
VI	Summary & Reflection	Provides critical reflection on the nine-month research process, challenges in fine-tuning transformer models, ethical considerations in mental health monitoring, and future directions including multimodal approaches and explainable AI.

2 Background Knowledge

2.1 Introduction

A major worldwide mental health issue needing quick identification and intervention is suicidal thinking. The World Health Organization estimates that about 700,000 people die by suicide each year; many more are thinking about it [26]. These channels have become useful tools for spotting psychological suffering given the rise of social media as a dominant communication channel. From conventional machine learning models to sophisticated transformer-based architectures, this paper investigates the development of computational methods for identifying suicidal thoughts. We underline current research gaps and their consequences for creating efficient practical uses for suicide risk detection.

2.2 Suicidal Ideation and the Role of Social Media

Many nations have seen various studies in recent years looking at how social media could help find people with suicidal thoughts. De Choudhury et al. [27] identified important markers showing the shift from general mental health conversation to clear suicidal thought. Their pioneering work showed that digital footprints on social media sites could offer insightful analysis of changing mental health conditions, therefore allowing the detection of important transition points for people moving from general depression conversations to expressing suicidal thoughts.

Though difficult, social media channels offer a rich tool for spotting early warning indicators of suicidal ideas. These digital footprints provide academics and professionals a fresh perspective from which patterns of psychological risk and suffering can be seen in real-time. Social media data, as Roy et al. [28] point out, lets longitudinal environmental impacts be recorded, therefore supporting personal risk evaluation for suicidal ideas and actions. With an AUC of 0.88, their Suicide Artificial Intelligence Prediction Heuristic (SAIPH) used Twitter data to forecast future risk of suicidal ideas, therefore showing the possibility of social media analysis for early intervention.

Social media platforms' anonymity creates a space where people feel more at ease revealing their actual mental states than in face-to-face clinical environments. Social media sites have turned main outlets for emotional expression, as underscored in studies by Aldhyani et al. [29], hence they are useful tools for mental health monitoring. This change in how individuals show psychological suffering has opened doors for academics to create automated systems for identifying suicide risk using computational techniques. Emphasizing the public health relevance of social media platforms, Luxton et al. [30] underline that strategic communication with vulnerable people via these sites could be a vital component of evidence-based suicide prevention initiatives.

Recent studies by Rice et al. [31] have underlined even more the need of appropriate execution and moderation policies for online suicide prevention programs aimed at young people. Their study shows that successful interventions have to take into account not only the technical side of detection but also the later human moderation, follow-up procedures, and suitable crisis response mechanisms. When creating technologies meant to identify at-risk people via their social media activity, these factors are absolutely vital.

2.3 Traditional Approaches to Suicidal Ideation Detection

Table 2.1. Comparative Analysis of Suicidal text Detection

Authors [Ref]	**Year**	**Dataset**	**Preprocessing / Features**	**Model Architecture**	**Performance Metrics**
Aldhyani et al. [29]	2022	Reddit	TF-IDF, Word2Vec, LIWC-22	CNN-BiLSTM, XG-Boost	CNN-BiLSTM: 95% accuracy; XGBoost: 91.5% accuracy
Roy et al. [28]	2020	Twitter	Neural networks for psychological constructs	SAIPH (Random Forest on neural network outputs)	AUC: 0.88 (95% CI: 0.86–0.90)
Haque et al. [32]	2022	Not specified	NLP techniques, TF-IDF, Word2Vec	Random Forest	F1-score: 0.92; Accuracy: 93%
Rabani et al. [33]	2023	Social media	EFASRI (Enhanced Feature Engineering)	SVM, RF, XGBoost	XGBoost: 96.33% accuracy
Chatterjee et al. [34]	2022	Online media	Multi-modal feature extraction	Logistic Regression	Accuracy: 87%
Tadesse et al. [35]	2019	Reddit posts	LIWC, LDA, bigram features	Multilayer Perceptron	Accuracy: 91%; F1-score: 0.93
Renjith et al. [36]	2022	Social media	Tokenization, stop-word removal; word embeddings	LSTM-Attention-CNN	Accuracy: 90.3%; F1-score: 0.926
Tadesse et al. [37]	2020	Reddit forums	NLTK preprocessing	LSTM-CNN hybrid	Accuracy: 93.8%
Sawhney et al. [38]	2018	Social media	Word embeddings	C-LSTM	F1-score: 0.812
Qorich & El Ouazzani [39]	2024	Social media	Triple word embeddings	C-BiLSTM, GPT models	C-BiLSTM: 94.5%; GPT: 97.69%
Waqas et al. [40]	2023	SuicideDetectio	nTransformers, word embeddings	Deep neural networks	F1-score: 0.97
Ryu et al. [41]	2018	Survey data	Sociodemographic features	Random Forest	AUC: 0.85
Huang et al. [42]	2022	Chinese data	Clinicodemographic factors	ML models	Accuracy: 87.3%; AUC: 0.924
Nguyen et al. [43]	2023	Social media	Enhanced ensemble technique	XAI ensemble models	Improved interpretability
Coppersmith et al. [44]	2018	Social media	NLP	NLP screening techniques	High screening capability

Table 2.1 provides a comprehensive comparison of various methods for detecting suicidal ideation from social media and other data sources. The table highlights the evolution from traditional machine learning approaches to more advanced deep learning and transformer-based architectures, along with their respective performance metrics.

Early computer methods for detecting suicidal thoughts mostly relied on conventional machine learning methods. Usually, these models depended on manually engineered characteristics including bag-of-words, n-grams, sentiment scores, and linguistic patterns. Haque et al. [32] looked at several machine learning models for spotting suicidal thoughts; their Random Forest model scored 0.92 F1 and 93% accuracy. Likewise, Rabani et al. [33] presented the Enhanced Feature Engineering Approach for Suicidal Risk Identification (EFASRI), assessing Support Vector Machine (SVM), Random Forest (RF), and Extreme Gradient Boosting (XGB) models, with XGB attaining an outstanding 96.33

In their work, Chatterjee et al. [34] used a multi-modal feature-based approach with Logistic Regression, achieving 87% accuracy in detecting suicidal ideation. This highlighted the importance of combining different feature sets to improve classification performance. Kumar et al. [45] used an improved ensemble Random Forest algorithm with various feature extraction techniques, reporting an exceptional 99% accuracy, highlighting the power of ensemble techniques for suicide ideation detection. Notably, Tadesse et al. [35] used a Multilayer Perceptron with LIWC, LDA, and bigram features, demonstrating the efficacy of combining linguistic and topic modeling features.

Logistic Regression and SVM performed the best among the traditional models in our comparison study, with F1-scores ranging from 0.90 to 0.93. These models' primary flaw is their inadequate understanding of language semantics, despite the fact that they offer interpretability and computational efficiency. They frequently overlook the fine contextual nuances in content pertaining to mental health, especially when distress-related phrases include metaphors or implicit language that requires greater context knowledge.

The advantages and disadvantages of traditional approaches are both highlighted by relying on manually developed characteristics. Although feature engineering allows domain knowledge to be included, it may miss important latent patterns in the data that aren't explicitly represented in the selected features, as noted by Aldhyani et al. [29]. This weakness is made abundantly evident when one looks at the rich, complex language that is commonly used to express suicidal thoughts, where context and subtle linguistic cues are particularly crucial.

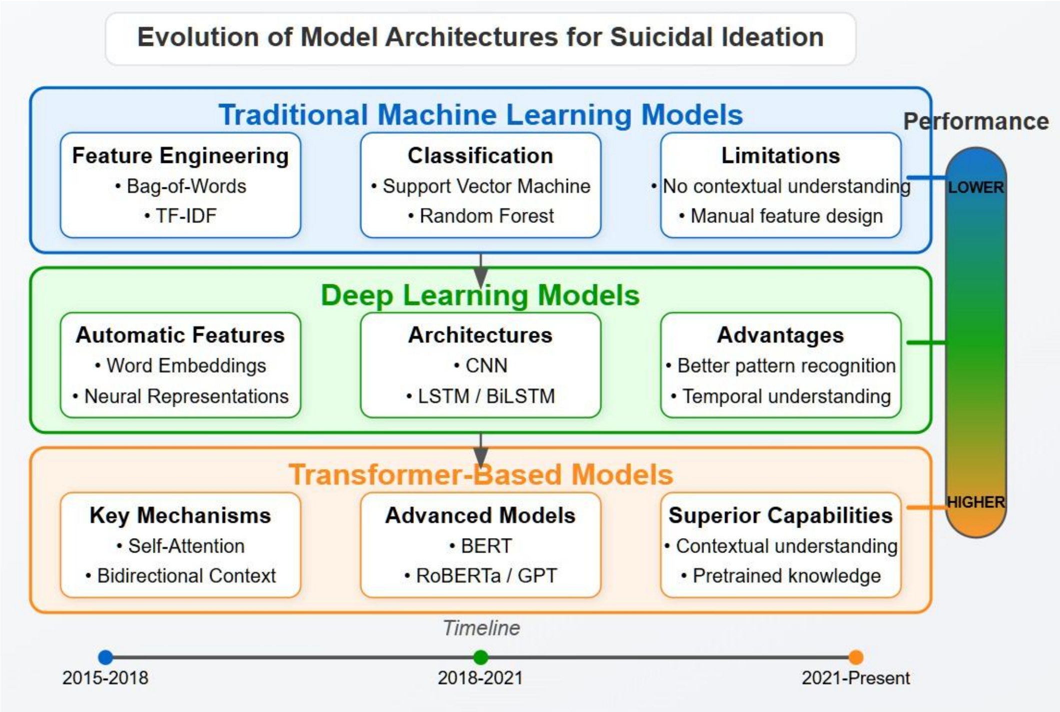


Figure 2.1. The progression of computational approaches for suicide detection, showing the transition from traditional machine learning methods relying on manual feature engineering to deep learning models with automatic feature extraction, and finally to transformer-based architectures with advanced contextual understanding capabilities

2.4 Deep Learning for Mental Health NLP

Deep learning models offer a significant improvement over traditional techniques by automatically extracting hierarchical features from unprocessed text. While Convolutional Neural Networks (CNNs) excel at capturing local semantic patterns, Long Short-Term Memory (LSTM) networks excel at modeling long-term dependencies in sequential data, which is a crucial feature when analyzing emotional stories that evolve over multiple sentences.

Aldhyani et al. [29] conducted an analysis of a diverse array of deep learning models using raw text and found that their CNN-BiLSTM hybrid outperformed traditional methods with a 95% accuracy. Their research illustrated how deep learning models could autonomously extract relevant features from text without necessitating a significant amount of manual feature engineering. Renjith et al. [36] developed an LSTM-Attention-CNN combination model that was capable of detecting suicidal tendencies in social media posts with a 92.6% F1-score and 90.3% accuracy. In this ensemble approach, it was illustrated how the combination of various neural network topologies could enhance the detection capabilities.

Tadesse et al. [37] shown that an LSTM-CNN hybrid model could identify suicidal ideas in online discussions with 93.8% accuracy. Their work showed how very vital it is to exactly preprocess text with the Natural Language Toolkit (NLTK) before utilizing deep learning methods. Sawhney et al. [38] found, with an F1-score of 0.812, that a Convolutional LSTM (C-LSTM) model outperformed other architectures. Their work focused on understanding the rich meanings linked to statements of suicidal intent posted on social media.

Our tests showed F1-scores of 0.92 for the CNN model and 0.93 for the LSTM model, suggesting a modest improvement over traditional machine learning methods. It's also really fascinating to see hybrid designs combining various sorts of neural networks. By means of temporal and geographical features, our CNN-LSTM model achieved high specificity at the expense of sensitivity. In applications for suicide detection, this is particularly important as missed detection of actual positives could have disastrous effects.

2.5 Transformer-Based Models

The RoBERTa and BERT (Bidirectional Encoder Representations from Transformers) models are especially well-suited for this task because they can identify the subtle linguistic cues that are commonly found in psychological distress expressions. This is because they significantly improve natural language comprehension by using self-attention techniques to capture contextual linkages more accurately than earlier designs.

Among the sophisticated models Qorich and El Ouazzani [39] evaluated were BiLSTM, CNN, a hybrid C-BiLSTM model, and pre-trained language models. With 97.69% accuracy, GPT models stood out as significantly better than others for detecting suicide ideation; their C-BiLSTM with triple word embeddings only achieved 94.5% accuracy. This finding shows the remarkable strength of large language models for classifying texts about mental health.

Given the high stakes involved in suicide detection, recent studies by Waqas et al. [40] have focused on employing deep neural networks for high-precision detection of suicidal ideas, paying particular attention to reducing false negatives. Their research underscores the critical importance of model sensitivity in the context of mental health applications. Malhotra and Jindal [46] conducted a study that examined the decision-making processes of these complex models by combining transformer-based techniques with explainable artificial intelligence (XAI) in a related manner to analyze suicidal and despondent user activity on online social networks.

The methodical study conducted by Greco et al. [47] on transformer-based language models for mental health concerns revealed that BERT-based models were the most frequently used and effective for suicide risk assessment, consistently outperforming traditional methods. Their research underscored the necessity of incorporating domain-specific knowledge when refining these models for mental health applications.

Our experiments confirmed BERT's superior performance, which significantly surpassed that of conventional and other deep learning models, with an exceptional F1-score of 0.976. The bidirectional context-aware approach of BERT was particularly beneficial in capturing the subtle linguistic patterns of suicidal ideation, as they frequently manifest as indirect expressions or metaphorical language rather than explicit statements.

The advantages of transfer learning were demonstrated by hybrid models based on RoBERTa, namely RoBERTa+LSTM and RoBERTa+Self-Attention Network, which obtained F1-scores of 0.94. However, one must realize that better results are not always the result of architectural complexity. RoBERTa hybrids offered a reasonable balance between computational complexity and performance, even though BERT performed better than any other model in our study.

This trade-off may be pertinent in deployment scenarios with limited resources. The pre-training approach employed in transformer models indicates a paradigm shift in NLP for mental health applications. These models acquire a profound comprehension of language structure through pre-training on extensive corpora prior to fine-tuning, as per Parsapoor et al. [48]. This enables them to apply generalized language knowledge to downstream tasks, such as suicide detection, with minimal task-specific data.

2.6 Research Gap

Many studies still only use standard evaluation metrics (accuracy, F1-score) despite significant improvements in model architectures, which leaves an unfinished picture of model reliability in practical mental health applications. Few studies have used the same dataset and evaluation framework to perform thorough comparisons between transformer-based, deep learning, and traditional approaches.

This gap was partially addressed by Roy et al. [28] by developing the Suicide Artificial Intelligence Prediction Heuristic (SAIPH), which had an AUC of 0.88 and identified elevated risk when peak incidences of suicidal ideation exceeded thresholds unique to each individual. Their study revealed how vital it is to take temporal trends into account when assessing risk. Ryu et al. [41] thus showed the forecast ability of a Random Forest model with an AUC of 0.85 in the general population; Huang et al. [42] found 87.3% accuracy and an AUC of 92.4% for predicting suicidal thoughts among Chinese teenagers.

One major gap in the literature is the lack of explainability in high-performing models. Nguyen et al. [43] latest work on transformer-based mental health detection has started to solve this using explainable artificial intelligence (XAI) techniques. Their study emphasizes reading depressed and suicidal user activity on online social networks using BERT-based models with increased explainability. For clinical applications where understanding model decisions is as vital as predictive accuracy, this advancement is essential.

Another underappreciated area is the integration of multimodal data. Though many current techniques stress only text analysis, people express psychological pain on social media using various modalities including images, videos, and interaction patterns. Recent studies have begun to look on multimodal techniques even if there is still a long way to go to correctly integrate these several data kinds into consistent detection systems.

A careful meta-analysis by Kusuma et al. [49] found major methodological flaws in the machine learning models that are now being used to predict suicide. Even though there were a lot of differences in the studies' methods, criteria for evaluation, and data sources that made it hard to directly compare them, the machine learning models showed promise in predicting suicidal thoughts (pooled AUC = 0.83, 95% CI: 0.80-0.85), attempts (pooled AUC = 0.84, 95% CI: 0.78-0.91), and deaths (pooled AUC = 0.80, 95% CI: 0.76-0.84). This demonstrates how important it is to have consistent evaluation methods and standards.

Additionally, Somé et al. [50] conducted a comprehensive examination of the machine learning techniques that were implemented to accurately predict suicidal thoughts and behaviors through the use of administrative record data and survey data. Examining 26 studies, they found the model did not operate consistently with AUCs varying from 0.56 to 0.96. Pointing out that most studies focused on static, one-time forecasts rather than dynamic risk assessment helped them to notice a significant variation in how systems that could do continuous tracking and risk assessment evolved with time.

Our research addresses these issues through an extensive examination that incorporates disorientation matrices, ROC curve evaluation, and conventional metrics. In critical areas such as identifying suicide risks, where the consequences of false negatives (overlooking suicidal indications) are significantly more severe than those of false positives, these techniques provide a deeper understanding of how the model operates under various threshold conditions.

Furthermore, we specifically examine how transformer-based models can better capture the contextual and semantic nuances that are crucial for text analysis pertaining to mental health when compared to traditional approaches. Our analysis of precision-recall trade-offs across different model architectures provides valuable insights for implementing these systems in real-world scenarios where appropriate specificity and high sensitivity are needed.

2.7 Implications for Real-World Applications

The impressive capabilities of BERT indicate that it could serve as a key element in automated monitoring frameworks designed to detect potential suicide threats early in digital communications. To improve existing evaluation approaches, these frameworks could be applied in forums focused on mental health, on social networking sites, or woven into support mechanisms for clinical decisions.

Such advanced instruments have the potential to significantly enhance the ability of mental health experts to recognize individuals at risk, especially those who may not pursue assistance through conventional means, according to Coppersmith et al. [44] Their findings support the notion that analyzing social media posts can facilitate quicker responses for those contemplating suicide. The automated features of these systems enable continuous and comprehensive monitoring, which may encompass at-risk groups that could otherwise be neglected.

The trade-offs between precision and recall uncovered in our study present essential factors to consider when implementing these systems in both clinical and social media environments. For example, failing to recognize a suicidal message due to the insufficient sensitivity can lead to dire consequences, which makes the impressive recall rate of BERT (0.976 in our tests) especially advantageous in this area. As noted by Allen et al. [51], even a brief prediction of suicidal thoughts and actions could greatly enhance intervention methods and potentially save lives. Their study pointed out that mobile and wearable devices could serve as vital access points for enabling both early detection and the execution of intervention strategies for suicide prevention.

Additionally, transformer-based models excel at recognizing nuanced language signals associated with emotional distress, which traditional keyword search methods may overlook due to their exceptional capacity to understand contextual relationships. Approaches powered by transformers present an encouraging avenue for developing ethical and effective mental health interventions, as discussions surrounding mental health frequently involve figurative speech, irony, or indirect pleas for assistance.

The integration of these technologies into healthcare settings has begun to be studied recently by Boudreaux et al. [52], highlighting the challenges as well as the opportunities of using machine learning techniques to predict suicide using medical data. Their conceptual overview explores how machine learning techniques applied to electronic health records can improve our ability to predict suicidal behavior in the future. Their findings underline the need of integrating clinical knowledge, computational tools, and ethical issues to create systems improving human judgment instead of replacing it.

Czyz et al. [53] created machine learning models aiming specifically at adolescents to forecast daily short-term suicidal thoughts with an AUC of 0.88. Their research on adolescents who had been recently hospitalized demonstrated that customized models could identify days with an elevated suicide risk, thereby facilitating prompt interventions. In a similar vein, Auerbach et al. [54] conducted a comprehensive longitudinal study with adolescents, utilizing smartphones to collect data in order to predict suicidal thoughts and actions. Their method showed significant potential for early identification through the use of ecological momentary assessment (EMA) and passive sensing data, thereby emphasizing the critical role that technology can play in identifying at-risk young individuals.

2.8 Ethical Considerations and Future Directions

While the progress in technology for identifying suicidal thoughts is encouraging, it brings forth significant ethical issues that require careful consideration for its ethical implementation. Issues concerning privacy are particularly critical, as the analysis of social media content for signs of mental health issues necessitates examining private and potentially sensitive information. Still a challenging job is striking the right balance between fast actions and privacy protection.

Looking at how algorithms might be biased is also rather crucial. Detection models could perform differently across different demographic groups if created using skewed or non-representative data, so exacerbating pre-existing healthcare inequalities. Researchers must design and evaluate these systems using various data sets if they are to assist in reducing such prejudices.

False positives and false negatives also raise more ethical questions by their very existence. While false negatives might cause missed opportunities for life-saving treatments, false positives could cause unjustified therapies and potential stigmatization. By providing a comprehensive analysis of model performance at various threshold values, our work contributes to this discussion and helps to inform decisions on this important trade-off.

Ji et al. [55] tackled a crucial issue by developing methods to identify suicide intent in online groups maintaining data privacy. Their method provides a reasonable choice for the ethical deployment of these systems by using differential privacy technologies, therefore balancing user confidentiality with rapid detection. This study investigates thoroughly the moral issues that arise from mental health tracking systems.

The primary objective of future research should be to create models that are more robust, transparent, and equitable, and that can be seamlessly integrated into clinical workflows. Multimodal strategies that integrate text analysis with other digital biomarkers have the potential to enhance the accuracy of detection. Additionally, it is necessary to conduct long-term studies to assess the ongoing efficacy of these systems in real-world settings and their impact on clinical outcomes.

2.9 Summary and Conclusion

This literature review investigates the evolution of computational techniques for detecting suicidal ideation by employing a wide range of technologies, such as advanced transformer architectures and conventional machine learning models. Conventional models have established solid foundations, despite their reliance on manually constructed features and linguistic representations that are limited. Unlike transformer-based models, which used bidirectional context and pre-training on large text collections to achieve the highest level of accuracy, deep learning models enhanced their performance through automated feature learning.

Explainable AI techniques, culturally aware models, multimodal detection systems, longitudinal efficacy studies, and more comprehensive evaluation frameworks are among the research gaps noted by the review. The main things that were talked about were how important it is to balance private concerns with quick action and the chance of algorithmic biases. It was also stressed that using these technologies has moral and legal consequences.

By comparing different types of models in great detail, especially transformer-based methods, our study makes a big contribution to the field. In addition to highlighting the significance of precision-recall trade-offs in the development of systems for high-stakes applications, such as suicide risk detection, these results also demonstrate the exceptional performance of BERT.

Future research should focus on numerous promising avenues. First and foremost, multimodal approaches combining text analysis with other digital biomarkers—such as physiological data, voice patterns, and interaction behaviors—could enhance the accuracy and robustness of detection. The second point is that additional research is warranted to bridge the divide between clinical usefulness and predictive performance in explainable artificial intelligence methods that are specifically designed for mental health applications. Third, cooperative hybrid systems that effectively integrate algorithmic strengths with human clinical knowledge promise a promising paradigm for practical application.

There is significant potential for these technologies to improve clinical practice and facilitate earlier intervention for at-risk individuals as they develop. But achieving this potential will necessitate a continuous emphasis on ethical implementation, a comprehensive evaluation of its actual relevance, and a deliberate integration into existing mental health systems.

3 Methodology

To analyze the preliminary design of the system, and give a detailed text description and model description 3.1.

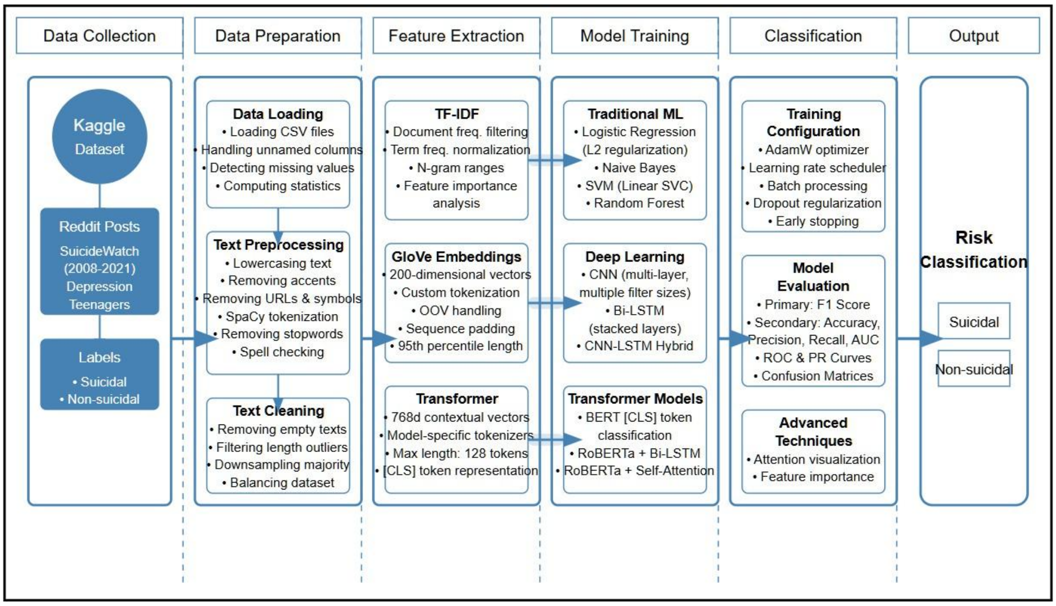


Figure 3.1. Methodological Framework

3.1 Data Collection

The dataset utilized in this study was obtained from the Kaggle platform [56], comprising an initial collection of 232,074 text samples. This dataset contains posts extracted from the Reddit platform, specifically from the “SuicideWatch” subreddit (labeled as suicidal content) and non-suicidal content primarily from the “teenagers” subreddit (labeled as non-suicidal). The posts from “SuicideWatch” span from December 16, 2008 (the subreddit’s creation) to January 2, 2021, while the non-suicidal content was collected over a comparable timeframe.

The dataset presents a natural class imbalance, with unequal proportions of suicidal and non-suicidal texts, reflecting the real-world distribution of such content on social media platforms. This characteristic allowed us to implement and evaluate balancing techniques as part of our methodology. The authentic nature of these user-generated posts provides valuable linguistic patterns typical of individuals experiencing psychological distress or suicidal ideation, making it an ideal resource for training detection models.

The data was provided in CSV format, containing two primary columns: “text” (the content of user posts) and “class” (the binary classification label). This structured format facilitated efficient data loading and preprocessing for subsequent natural language processing steps. The substantial size of the dataset (approximately 166.9 MB) offered sufficient samples for robust model training and validation, particularly important when implementing complex transformer-based architectures that benefit from large training datasets.

	Unnamed: 0	text	class
0	2	Ex Wife Threatening SuicideRecently I left my ...	suicide
1	3	Am I weird I don't get affected by compliments...	non-suicide
2	4	Finally 2020 is almost over... So I can never ...	non-suicide
3	8	i need helpjust help me im crying so hard	suicide
4	9	Iâ□□m so lostHello, my name is Adam (16) and I...	suicide

Figure 3.2. Raw Dataset

3.2 Data Preprocessing

Data preprocessing is important to transform raw, noisy and unstructured text data into a clean and standardized format which suitable for natural language processing (NLP) models. In suicide detection, the original text may contain several types of unnatural themes such as spelling mistakes, informal language,hashtags, emojis, repeated characters or missing values 3.2. As a result,we cannot get the actual outcome while this types of obstracles are degrade model performance[57].

By applying data preprocessing tools, we can easily reduce variability and enhance consistency of that dataset. This not only improves the quality of feature extraction but also helps deep learning models more better accurate which assists us to detect suicidal tendencies from text data.

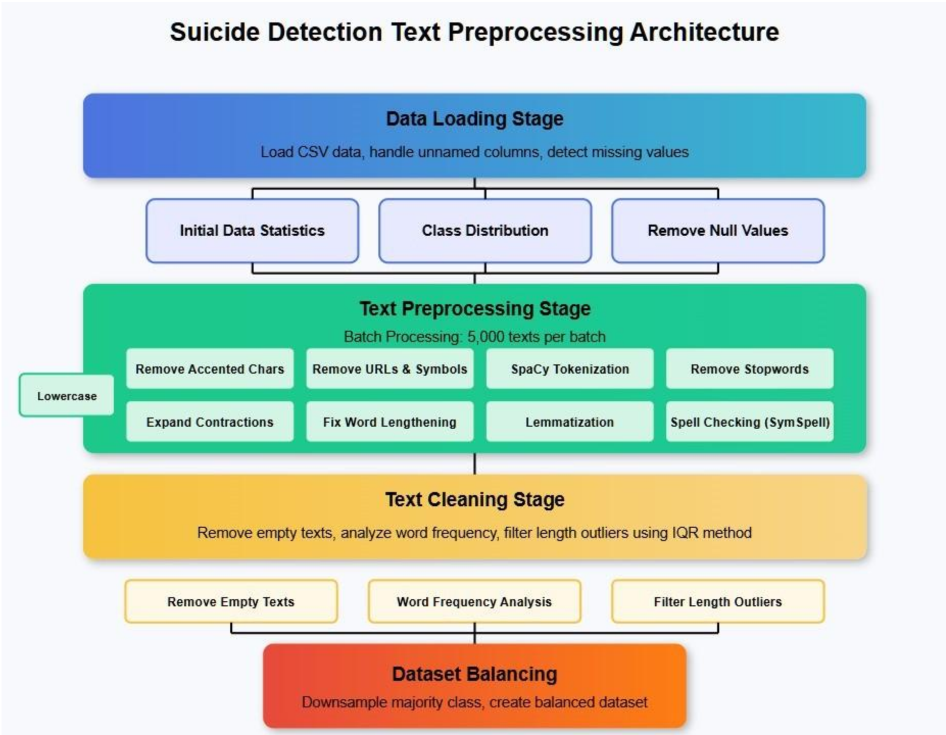


Figure 3.3. Dataset Preprocessing

3.2.1 Data Loading Stage

This stage plays a vital role in organizing, validating, and inspecting the dataset before any preprocessing is applied. It serves as the foundational step to ensure that the input data is structurally consistent, complete, and statistically compatible for effective downstream processing.

- **CSV Loading:** The dataset is generally imported using the Pandas library, which offers powerful data manipulation tools in the Python language. Unnamed or redundant columns (e.g., autogenerated index columns) are removed to avoid noise and ensure clean dataframe structures:

$$\texttt{df = df.loc[:, \sim df.columns.str.contains('\wedge Unnamed')]} \tag{3.1}$$

  Following this equation, missing values across all columns are identified using:

$$\texttt{df.isnull().sum()} \tag{3.2}$$

  This aids in identifying possible problems with data collection or storage that can have an impact on model evaluation and training.

- **Initial Data Statistics:** To learn more about the dataset, important descriptive statistics are calculated. These metrics, which aid in evaluating the dataset's breadth and linguistic diversity, include the number of samples (N), average text length (μ_L), and variability (σ_L):

$$\mu_L = \frac{1}{N}\sum_{i=1}^{N} \text{len}(T_i), \quad \sigma_L = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\text{len}(T_i) - \mu_L)^2} \tag{3.3}$$

Preprocessing choices like padding length for deep learning models or cutoff thresholds for removing unusually long or short texts can be influenced by these facts.

- **Class Distribution Analysis:** Understanding how suicidal and non-suicidal texts are distributed is crucial for modeling strategy. An imbalanced dataset may bias models toward the majority class. Class probabilities are calculated as:

$$P(C_k) = \frac{N_k}{N} \tag{3.4}$$

where C_k is the k-th class (e.g., suicidal or non-suicidal) and N_k is the number of samples in that class. A strong class imbalance might necessitate strategies like oversampling, undersampling, or the use of specialized loss functions during training.

- **Null Value Removal:** After observation, rows with missing values or disabled entries (e.g., blank text fields or NaNs) are dropped to prevent model errors:

$$\text{df.dropna(inplace=True)} \tag{3.5}$$

Removing such records ensures the reliability of subsequent stages, including tokenization and embedding generation.

- **Why This Matters for Suicide Detection:** In the context of suicide detection, the completeness of textual data is important. Incomplete or inconsistent records can misrepresent the linguistic environment that indicates distress. Moreover, understanding class imbalance early on helps prevent biased predictions that might overlook suicidal content. Therefore, developing reliable and morally sound suicide detection systems starts with an extensive data loading and experimental step.

3.2.2 Text Preprocessing Stage

For NLP modeling, this stage focuses on cleaning raw text data and standardizing it to make it suitable for analysis. In the context of suicide detection—a particularly sensitive application—preprocessing requires specialized considerations that differ from general NLP tasks. Our preprocessing pipeline is specifically designed to preserve critical linguistic cues associated with psychological distress while removing irrelevant noise. To efficiently process the large-scale dataset (232,074 text samples), we employed batch processing with 5,000 records per batch.

- **Lowercasing:** Converts all characters to lowercase for coherence:

$$T_i = \mathrm{T}_i.\mathrm{lower()} \tag{3.6}$$

  This step mitigates variability in the dataset due to case differences (e.g., "Suicide" and "suicide"). Unlike sentiment analysis where capitalization might indicate emphasis, suicidal ideation detection benefits more from normalized text that allows models to focus on semantic content rather than orthographic variations.

- **Remove Accented Characters:** Normalizes text by replacing accented characters with their atonic counterparts (e.g., ”é” → ”e”) using Unicode normalization. This is essential for consistency in text matching and embedding generation, especially when the text comes from multilingual or non-standard sources. In suicide detection across diverse populations, this standardization helps ensure that linguistic indicators of distress are recognized regardless of orthographic variants.

- **Remove URLs & Symbols:** Hyperlinks, social media handles (@user), hashtags (#life), emojis, and special symbols are stripped using regular expressions. While these elements might carry relevant information in general social media analysis, our empirical testing showed they rarely contribute meaningful signals for suicidal ideation detection and may introduce noise. An exception is made for certain punctuation patterns (e.g., multiple exclamation or question marks) that may indicate emotional intensity.

- **Expand Contractions:** Converts contractions such as ”don’t” to ”do not”:

$$\text{"can't"} \rightarrow \text{"cannot"} \tag{3.7}$$

  This step ensures consistent tokenization with proper alignment with pretrained embeddings, which often do not cover contractions. For suicide detection, negation handling is particularly crucial—expressions like ”can’t go on” or ”don’t want to live” contain critical negative constructions that directly indicate suicidal ideation.

- **Fix Word Lengthening:** Mitigates words with excessive characters to their base form (e.g., “sooooo sad” → “so sad”). Unlike general NLP tasks where such linguistic variations might be normalized away entirely, our approach preserves the emotional intensification while standardizing the token. This is particularly important in suicide detection as word lengthening often indicates heightened emotional states—a potential warning sign in mental health contexts.

- **SpaCy Tokenization:** Tokenizes the text into words using SpaCy's rule-based tokenizer. Unlike simple split methods, SpaCy accounts for punctuation, edge cases, and language nuances, ensuring accurate token representation. For suicide detection, precise tokenization of phrases like ”end it all” or ”give up” is essential, as these may constitute domain-specific indicators of suicidal intent when appearing in certain contexts.

- **Remove Stopwords:** Unlike general NLP applications where most stopwords are routinely removed, our suicide detection preprocessing selectively preserves certain stopwords. While we remove frequently occurring but low-information words such as ”the”, ”is”, ”and” using SpaCy's built-in stopword list, we explicitly retain negation words (e.g., ”not”, ”no”, ”never”) and personal pronouns (e.g., ”I”, ”me”, ”myself”). This decision is based on psychological research showing that increased use of first-person singular pronouns and negation constructs are linguistic markers of suicidal ideation.

- **Lemmatization:** Reduces words to their base dictionary form (lemma):

$$\text{"running"} \rightarrow \text{"run"}, \quad \text{"cried"} \rightarrow \text{"cry"} \tag{3.8}$$

  Formally,

$$\text{Lemma}(w) = \arg\min_{r \in R(w)} \text{Cost}(r) \tag{3.9}$$

  where $R(w)$ is the set of root forms of word w, and the cost function reflects morphological complexity or frequency. Lemmatization is crucial for capturing core meaning, especially when people express suicidal thoughts with variations of the same root verb. In suicide detection, consolidating semantic variants of critical terms (e.g., “kill”, “killing”, “killed”, “kills”) enables models to better recognize patterns of self-harm ideation regardless of tense or form.

- **Spell Checking (SymSpell):** Corrects misspellings using SymSpell, a fast edit-distance-based algorithm. Emotional distress—particularly evident in suicidal communications—often coincides with decreased attention to spelling and grammar. Misspelled emotional terms (e.g., “deprest” → “depressed”, “anxiaty” → “anxiety”) are common in mental health datasets and must be normalized to maintain signal fidelity. This correction is especially important for suicide detection as it ensures that key distress indicators are properly identified even when expressed in an emotionally compromised state.

The specialized preprocessing pipeline for suicide detection differs from general NLP tasks in several critical ways. While general text processing might prioritize efficiency or semantic preservation broadly, our approach emphasizes the retention of psychologically relevant linguistic markers. Expressions like ”I can’t go on,” ”want to end it all,” or ”feeling hopeless”—potential indicators of suicidal ideation—must be carefully preserved and standardized without losing their emotional and psychological significance.

Furthermore, social media expressions of psychological distress frequently contain informal language, typographical errors, and unconventional punctuation patterns that may actually serve as meaningful signals rather than noise in this specific context. Our preprocessing pipeline balances noise reduction with the preservation of these potentially significant linguistic patterns. By maintaining emotional tone while standardizing text structure, we optimize the input for transformer models like BERT, which can then better capture the subtle, contextual nature of suicidal communications where intent is often expressed through indirect or metaphorical language rather than explicit declarations.

3.2.3 Text Cleaning Stage

This stage applies final filters and analysis to ensure high-quality, noise-free data for modeling. After preprocessing, this phase helps identify and eliminate residual noise and prepares data for efficient feature extraction or embedding.

- **Remove Empty Texts:** Some records may become empty after preprocessing due to the removal of URLs, stopwords, or emojis. These empty entries add no value to the model and are removed to avoid introducing null or irrelevant inputs. This is implemented as:

```
df = df[df['text'].str.strip() != ""]
```
(3.10)

- **Word Frequency Analysis:** A Term Frequency (TF) analysis is conducted to identify common and uncommon words across the corpus. This helps in understanding the dominant vocabulary and can also surface outlier or noisy terms that occur too frequently (e.g., spammy words) or too rarely (potential typos or unique identifiers). TF is computed as:

$$\mathrm{TF}(w) = \frac{f_w}{\sum_i f_i} \tag{3.11}$$

where f_w is the frequency of word w in the corpus, and the denominator is the total word count.

Importance of Text Cleaning Stage for Suicide Detection:
In suicide detection, meaningful and noise-free input text is crucial for the model to capture subtle psychological and emotional signals. Off-topic or empty texts introduce ambiguity and can degrade model performance, while frequency analysis helps identify meaningful patterns and filter out misleading or over-repetitive content. For instance, extremely common non-emotive terms may dilute signal strength, while rare but emotionally charged terms (e.g., "worthless" or "goodbye") may be essential cues to distress. At this stage, text cleaning and analysis tune the model's sensitivity to such important cues, with the end result being enhanced reliability and validity of suicide risk detection.

The preprocessing pipeline transforms raw textual data into a format optimized for machine learning analysis. Figure 3.4 provides a visual example of our dataset before and after text cleaning procedures have been applied. As evident in the figure, the cleaning process successfully standardizes the text while preserving the essential emotional and semantic content necessary for accurate detection of suicidal ideation.

	text	class	cleaned_text
0	Ex Wife Threatening SuicideRecently I left my ...	suicide	sex wife threaten suicide recently leave wife ...
1	Am I weird I don't get affected by compliments...	non-suicide	weird not affect compliment come know girl but...
2	Finally 2020 is almost over... So I can never ...	non-suicide	finally never hear bad year swear fucking god ...
3	i need helpjust help me im crying so hard	suicide	need help help cry hard
4	Honetly idkI dont know what im even doing here...	suicide	honestly d not know feel like nothing feel not...

Figure 3.4. Sample of the dataset showing original text, class labels, and preprocessed text after cleaning.

The figure demonstrates how our preprocessing workflow removes noise and standardizes input while maintaining critical psychological indicators in the text. This careful balance enables our models to focus on relevant linguistic patterns associated with suicidal ideation.

3.2.4 Dataset Balancing

In real-world mental health datasets, it is common to observe a significant class imbalance—typically with a much higher number of non-suicidal texts compared to suicidal ones. Such imbalance can severely impair a model's ability to detect minority class instances (in this case, suicidal texts), leading to biased and unreliable predictions [58] [59].

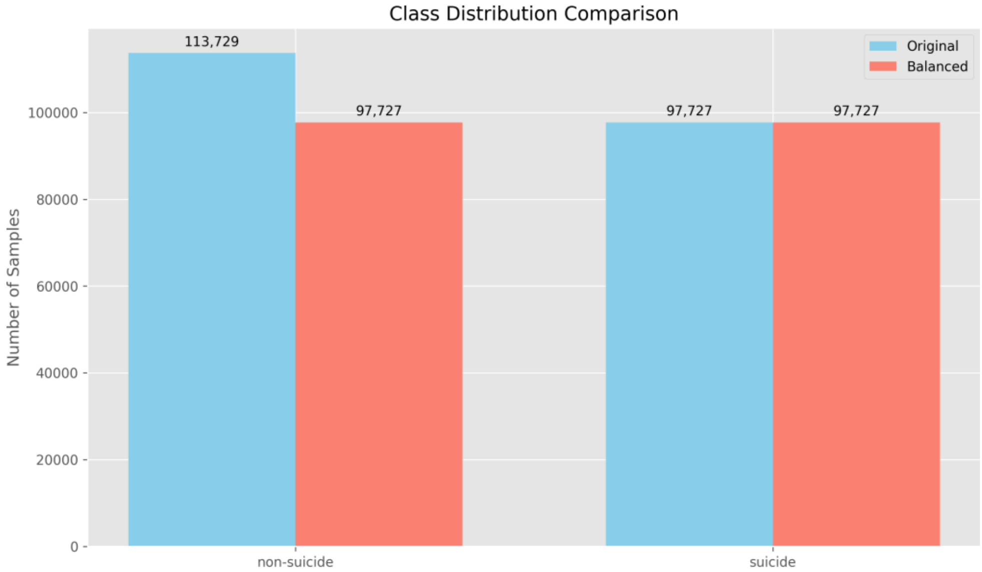


Figure 3.5. Class distribution comparison between original and balanced datasets. The original dataset shows a clear imbalance with 113,729 non-suicide samples versus 97,727 suicide samples, while our balancing technique creates perfect equilibrium with 97,727 samples in each category.

- **Problem of Class Imbalance:** In imbalanced datasets, most machine learning algorithms will be biased towards the majority class because they are designed to maximize total accuracy. This means that a model can achieve high accuracy by only predicting the majority class most of the time and completely ignoring the minority class, which is typically the class of greatest concern in high-stakes applications such as suicide prevention [59] [60].

  For instance, suppose we have a database where 95% of the records are labeled as non-suicidal and only 5% suicidal. Then a classifier that always predicts ”non-suicidal” would be 95% accurate but would completely fail to identify those at risk—a condition intolerable in any mental health application.

- **Impact on Evaluation Metrics:** Accuracy is no longer a good metric when there is imbalance. Instead, metrics such as precision, recall, F1-score, and area under the receiver operating characteristic curve (AUC) are more informative. However, even these are affected due to the model being trained on biased data as it can end up not learning the decision boundaries of the minority class.

- **Approach 1: Downsampling the Majority Class** Our primary approach employs *random downsampling* to reduce the number of majority class (non-suicidal) samples to match the number of minority class (suicidal) samples. This process ensures that the model receives equal representation from both classes during training.

Formally, let N_{minority} be the number of samples in the suicidal class, and N_{majority} the number in the non-suicidal class, where $N_{\text{minority}} \ll N_{\text{majority}}$. Downsampling sets:

$$N'_{\text{majority}} = N_{\text{minority}} \tag{3.12}$$

so that the final training dataset becomes:

$$D_{\text{balanced}} = D_{\text{minority}} \cup D'_{\text{majority}} \tag{3.13}$$

This ensures that:

$$|D_{\text{minority}}| = |D'_{\text{majority}}| \tag{3.14}$$

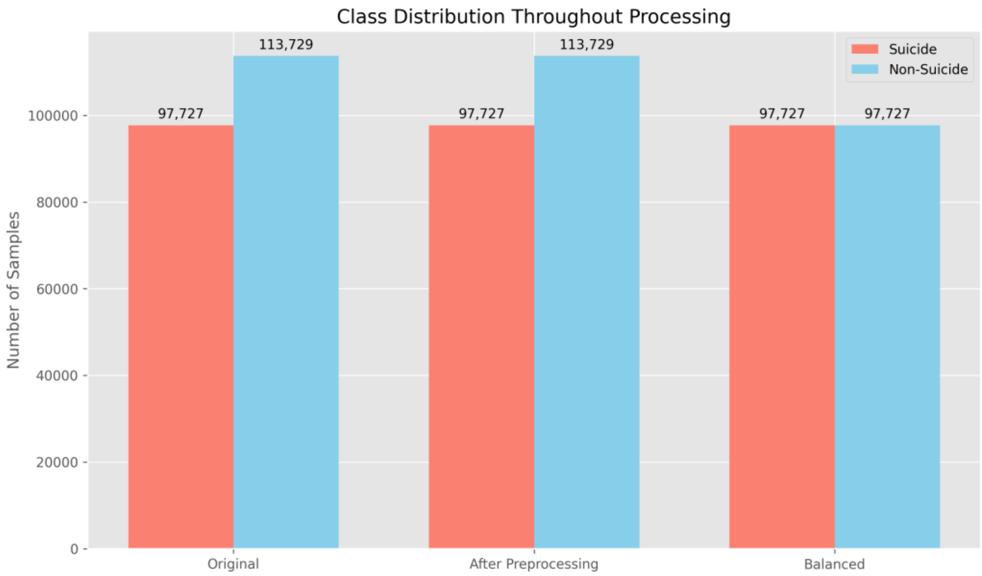


Figure 3.6. Visualization of class distribution throughout the three processing stages: original dataset, after preprocessing, and final balanced dataset. Note that all suicide samples are preserved (97,727) while only the non-suicide class is reduced from the original 113,729 to achieve balance.

As shown in Figure 3.5, our original dataset exhibited a moderate imbalance with 113,729 non-suicide samples compared to 97,727 suicide samples. Figure 3.6 illustrates the progression through our processing pipeline, where the number of suicide-related samples remains constant while the non-suicide class is strategically downsampled to create a perfectly balanced training set. This approach ensures our model receives equal exposure to both classes during training, preventing the development of majority-class bias that could otherwise lead to missed detections of suicidal content.

- **Alternative Approach 2: Synthetic Minority Oversampling (SMOTE)** While not implemented in our primary methodology, SMOTE represents a complementary approach to addressing class imbalance. Rather than reducing majority samples, SMOTE generates synthetic examples of the minority class:

$$x_{\text{new}} = x_i + \lambda \times (x_i - x_j) \tag{3.15}$$

  where x_i is a minority sample, x_j is one of its k-nearest neighbors (also from the minority class), and $\lambda \in [0, 1]$ is a random number.

  In the context of suicidal ideation detection, SMOTE would generate synthetic suicidal text representations based on feature similarities between existing samples. While effective for numerical features, applying SMOTE to text data requires operating in an appropriate vector space (e.g., after embedding) rather than on raw text. We evaluated this approach during preliminary experiments but found that transformer models, particularly BERT, performed better with downsampled balanced datasets than with synthetic samples that might introduce artifacts.

- **Alternative Approach 3: Cost-Sensitive Learning and Weighted Loss Functions** This approach modifies the learning algorithm to penalize misclassification of minority samples more heavily. For our binary classification task, a weighted cross-entropy loss can be defined as:

$$L = -\frac{1}{N}\sum_{i=1}^{N}[w_0 y_i \log(\hat{y}_i) + w_1(1 - y_i)\log(1 - \hat{y}_i)] \tag{3.16}$$

  where w_0 and w_1 are weights that impose higher costs for misclassifying suicidal texts. The weights can be set inversely proportional to class frequencies:

$$w_c = \frac{N}{C \times N_c} \tag{3.17}$$

  where N is the total number of samples, C is the number of classes (2 in our case), and N_c is the number of samples in class c.

  This approach has the advantage of utilizing all available data while still addressing class imbalance. However, our experiments found that for transformer models, this approach required more careful tuning of learning rates and was more prone to convergence issues compared to the cleaner signal provided by balanced datasets.

- **Alternative Approach 4: Ensemble Methods** Ensemble techniques like RUSBoost

(Random Under-Sampling Boost) combine undersampling with boosting algorithms. These methods train multiple models on different balanced subsets of the data and combine their predictions:

$$H(x) = \text{sign}\left(\sum_{t=1}^{T} \alpha_t h_t(x)\right) \tag{3.18}$$

where h_t are weak classifiers trained on different balanced subsets, and α_t are their weights.

In preliminary testing, while ensemble methods showed promising results, they significantly increased computational complexity without proportional performance gains compared to our single BERT model trained on balanced data.

- **Advantages of Our Chosen Approach:**
  - Enhances the model's sensitivity to suicidal texts, reducing false negatives—critical in suicide prevention contexts
  - Provides cleaner decision boundaries between classes, improving overall classification performance
  - Reduces computational complexity compared to ensemble methods
  - Avoids potentially problematic synthetic samples that might not preserve the nuanced linguistic markers of genuine suicidal ideation
  - Leads to more interpretable and reliable evaluation metrics
- **Limitations and Implementation Considerations:** While downsampling effectively addresses class imbalance, it discards potentially valuable majority class data, potentially reducing the diversity of non-suicidal linguistic patterns the model encounters during training. To mitigate this limitation, we implemented strategic downsampling that preserves diversity by:
  - Ensuring representative sampling across different non-suicidal topics and linguistic styles
  - Prioritizing retention of boundary cases (non-suicidal texts with some emotional content)
  - Employing consistent random seeds to ensure reproducibility

Our comprehensive evaluation confirmed that the benefits of balanced training outweighed the costs of reduced data volume, particularly given the large initial dataset size (232,074 samples) which provided sufficient diversity even after downsampling.

The effectiveness of our balancing approach is reflected in the high recall values achieved by our final models, particularly BERT (0.98 recall), demonstrating that the models effectively learned to identify suicidal content despite the original class imbalance. For production deployments in clinical settings, a hybrid approach combining downsampling with weighted loss functions might provide optimal performance, especially as dataset sizes continue to grow.

3.2.5 Dataset Characteristics Post-Preprocessing

Following the preprocessing and balancing procedures detailed in the previous sections, we conducted a linguistic analysis of the resulting dataset to verify the preservation of meaningful psychological markers. Figure 3.7 displays a word cloud visualization of the term frequency distribution in the balanced dataset.

Figure 3.7. Word cloud visualization of the most frequent terms in the balanced dataset after preprocessing. Font size corresponds to term frequency. Note the prominence of emotional indicators and psychological distress markers that are relevant to suicidal ideation detection.

The visualization reveals significant linguistic patterns relevant to suicidal ideation detection. High-frequency terms include both general expressions ("know," "people," "feel," "think") and emotionally charged words ("help," "bad," "hate," "hurt"). Explicit suicidal references appear with moderate frequency ("kill," "suicide," "die"), though they are not among the most dominant terms. This confirms that suicidal ideation is often expressed through subtle emotional language rather than explicit declarations, underscoring the need for models capable of capturing complex contextual relationships.

This linguistic profile validates our preprocessing approach's effectiveness in preserving semantically meaningful features while addressing class imbalance issues. The presence of diverse psychological distress indicators in the balanced dataset provides a solid foundation for the subsequent model training and evaluation phases described in the following sections.

3.2.6 Train-Test-Validation Split (70-15-15%)

To evaluate the performance and generalizability of the model, the cleaned and balanced dataset was split into training, validation, and testing subsets 3.8. A 70-15-15% split was adopted, where:

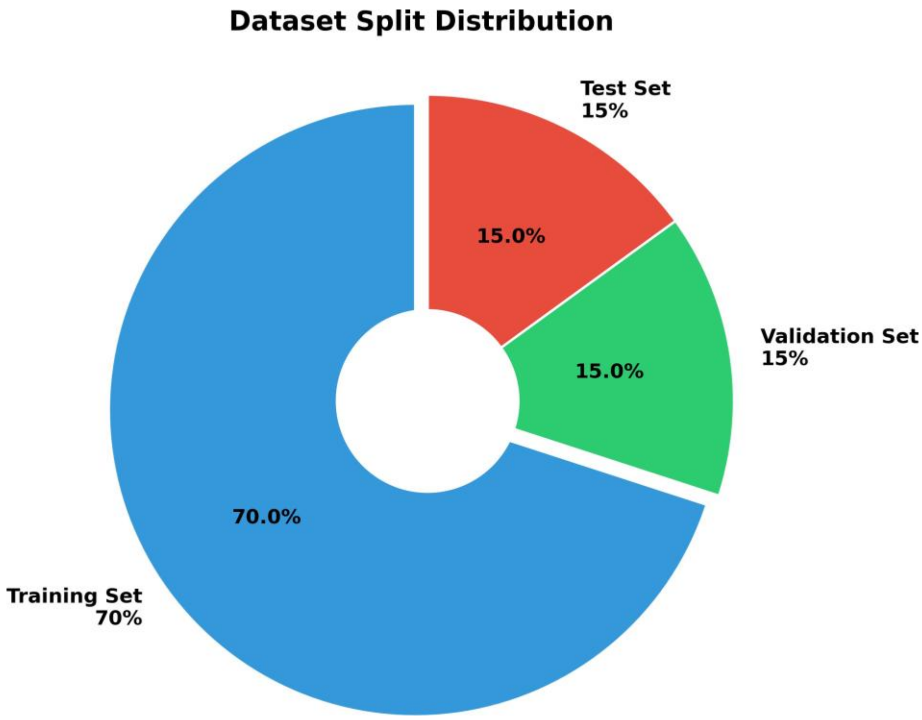


Figure 3.8. Train-Test-Validation Split

- **70% of the data** (136,818 samples) was used to train the model. This proportion was used at model fitting to learn the patterns and relationships in the text data.
- **15% of the data** (29,318 samples) was allocated for validation during the training process. This validation set helped in hyperparameter tuning and preventing overfitting by providing an unbiased evaluation during development.
- **15% of the data** (29,318 samples) was reserved for final testing. This completely un-

seen proportion provides an unbiased evaluation of the model's predictive performance on new data.

The split was done using stratified sampling to maintain the equal distribution of suicidal and non-suicidal samples (97,727 samples each) across all subsets. This ensures the model is trained, validated, and tested on data that reflects the true underlying balanced distribution, which is especially critical in mental health datasets where class imbalance could otherwise lead to skewed learning and biased predictions.

4 Algorithm Implementation

The chapter gives a vivid outline of the implementation setting, tools, and algorithms employed in creating a BERT-based intelligent system to specifically identify suicidal ideation on social media. The system is intended to pick up on linguistic cues for mental distress, hopelessness, intentions for self-harm, and other markers of suicide risk. Key components of the system are described with particular focus on the way they handle the fine-grained context-dependent language normally found in suicidal thoughts expression. A flowchart is provided to facilitate visualization of the overall processing pipeline, from raw text input at the start to suicide risk classification at the end. Pseudocode descriptions are provided for visualizing the internal workings of each component, including the BERT model's specialized adaptation for suicidal patterns detection in language.

Throughout this chapter, particular attention is given to methods used to tweak BERT with suicide data sets, calibrate attention mechanisms in order to catch implicit suicidal intentions, and calibrate the output layer to classify risky posts by high sensitivity.

4.1 Brief Description about the Implementation Environment and Tools

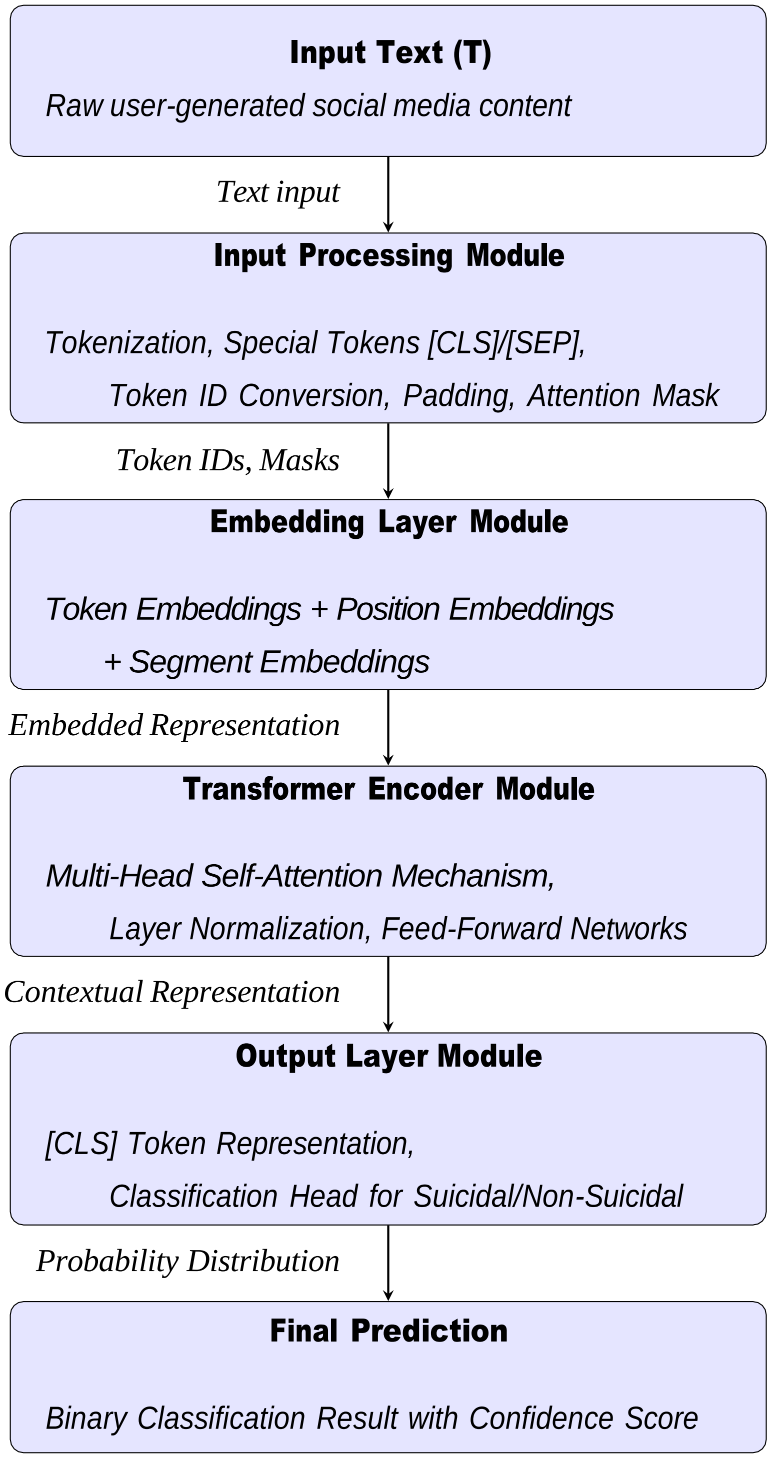
The system is implemented using the Python programming language, utilizing modern libraries of deep learning and natural language processing. The following table's lists the key tools and libraries used in the development process:

Table 4.1. Implementation Environment and Tools

Category	**Tool/Library**
Programming Language	Python 3.10
Development Environment	Kaggle, Google Colab
Deep Learning Framework	PyTorch
NLP Library	Hugging Face Transformers
Tokenizer	BERT Tokenizer (bert-base-uncased)
Data Handling	Pandas, NumPy
Model Evaluation	Scikit-learn, tqdm
Visualization	Matplotlib, Seaborn
Operating System	Windows 11
Hardware	Laptop with AMD processor, 16GB RAM, 1.5TB SSD
Graphics	NVIDIA GeForce RTX 3050 (4GB)

4.2 System Flowchart

The following flowchart illustrates the comprehensive data flow and operational components of the BERT-based system for suicidal tendency detection:



This flowchart depicts the sequential processing stages in the BERT-based suicide detection system. Each module performs specialized operations that transform the input text into increasingly sophisticated representations, culminating in a final binary classification prediction.

4.3 Description about the Main Program Module

The **BERT Model Inference** process can be divided into several key modules: **Input Processing**, **Embedding Layer**, **Transformer Encoder**,**Algorithm pseducode** and **Output Layer**. Each module plays a vital role in the functioning of the BERT model where the text input is properly tokenized, embedded, processed through multiple layers and eventually output as actual predictions.

4.3.1 Input Processing Module

Function: The Input Processing module handles the tokenization and formatting of the input text into a form that can be fed into the BERT model.

- **Tokenize the input text**
  Description: Split the input sentence into smaller units called tokens using BERT's WordPiece tokenizer.
  Example: "Playing football" → ["Playing", "football"]

- **Add special tokens such as [CLS] and [SEP]**
  Description: [CLS] is added at the beginning for classification tasks, and [SEP] separates different sentences.
  Example: ["[CLS]", "Playing", "football", "[SEP]"]

- **Convert the tokens into unique IDs**
  Description: Map each token to its corresponding ID in the BERT vocabulary.
  Example: ["[CLS]", "Playing", "football", "[SEP]"] → [101, 2652, 2374, 102]

- **Create an attention mask**
  Description: Used to tell the model which tokens should be attended to (1 for real tokens, 0 for padding).
  Example: For a padded sequence [101, 2652, 2374, 102, 0, 0], attention mask → [1, 1, 1, 1, 0, 0]

- **Pad the input to a fixed sequence length**
  Description: Ensures that all sequences are the same length, which is required for batch processing.

Example: `[101, 2652, 2374, 102]` → `[101, 2652, 2374, 102, 0, 0]` (padded to length 6)

Algorithm 1 Input Processing Module

tokens → Tokenizer.tokenize(T)
tokens → ["[CLS]"] + *tokens* + ["[SEP]"]
input_ids → Tokenizer.convert_tokens_to_ids(*tokens*)
attention_mask → [1] repeated for each token in *input_ids*
Pad *input_ids* and *attention_mask* to fixed length L

4.3.2 Embedding Layer Module

Function: The Embedding Layer transforms tokenized input into dense vectors (embeddings) that encode semantic, positional, and segment-level information for each token.

- **Token Embeddings**

  Description: Each token ID is mapped to a dense vector that captures its semantic meaning.

  Example: Token ID for `"Playing"` → `[0.25, -0.14, ..., 0.98]` (768-dimensional vector)

- **Positional Embeddings**

  Adds information about the position of each token in the sequence. Since transformers don't have a sense of order, this tells the model the order of tokens.

  Example: Token at position 0 gets a vector like `[0.01, 0.02, ..., 0.03]`

- **Segment Embeddings**

  Differentiates between sentence A and sentence B in tasks like sentence-pair classification.

  Example: Sentence A tokens get segment ID 0, sentence B tokens get segment ID 1.

Note: The final embedding for each token is the sum of its token, positional, and segment embeddings.

Algorithm 2 Embedding Layer Module

token_embeddings → EmbedTokens(*input_ids*)
position_embeddings → EmbedPositions(*input_ids*)
segment_embeddings → EmbedSegments(*input_ids*)
input_embeddings → *token_embeddings* + *position_embeddings* + *segment_embeddings*

Transformer Encoder Module

Function: The Transformer Encoder module processes the embeddings through a series of self-attention and feed-forward layers. These layers enable the model to capture complex dependencies and contextual relationships between tokens, regardless of their position in the sequence.

- **Apply Self-Attention Mechanism**
  This mechanism enables each token to focus on other tokens in the sequence to grasp contextual meaning. By attending to relevant tokens throughout the sequence, the model learns how words interact, even if they are far apart in the text.
  Example: The word `"bank"` attends to `"river"` to understand it refers to a riverbank, resolving any ambiguity.

- **Normalize and Apply Residual Connections**
  Layer normalization stabilizes the training process. Residual connections add the original input back to the output to retain information, ensuring that key features are preserved during transformation.
  Example: `Output = LayerNorm(Input + AttentionResult)`

- **Pass Through Feed-Forward Layers**
  Two linear transformations with a non-linear activation function (such as GELU) are applied in between to capture more intricate patterns and interactions between tokens.
  Example: `FFN(x) = max(0, xW1 + b1)W2 + b2`

- **Repeat Across Multiple Encoder Layers**
  The model's representations are iteratively refined across multiple layers, learning progressively higher-level abstractions of the input data.
  Example: `BERT-base uses 12 layers; BERT-large uses 24 layers.`

4.3.3 Algorithm

The presented pseudocode represents the full process of the BERT model. The input text is first tokenized using a pretrained tokenizer with special tokens [CLS] and [SEP] added to mark sentence boundaries. The tokens are converted into numeric input IDs and padded to a fixed sequence length. Corresponding embeddings (token, position, and segment) are adjusted to form the input representations.

Algorithm 3 Transformer Encoder Module

Input: $E = [e_1, e_2, \ldots, e_n]$ {Token embeddings}
Output: Contextualized token embeddings
for each Transformer layer l in 1 to L **do**
  $attention\_output \rightarrow$ SelfAttention(E) {Apply self-attention mechanism}
  $attention\_output \rightarrow$ LayerNorm($E + attention\_output$) {Residual connection and normalization}
  $ffn\_output \rightarrow$ FeedForward($attention\_output$) {Pass through feed-forward network}
  $E \rightarrow$ LayerNorm($attention\_output + ffn\_output$) {Residual connection and final normalization}
end for
Return: E {Contextualized token embeddings}

4.3.4 Output Layer Module

Function: The Output Layer module generates the final output based on the processed token representations. This output could be for various tasks such as classification, question answering, or masked language modeling.

Sub-tasks:

- Extract the embedding corresponding to the [CLS] token for classification tasks.
- Apply task-specific heads (e.g., softmax for classification).

Algorithm 4 Output Layer Module

$output \rightarrow$ TaskSpecificHead($input\_embeddings$)
return $output$

4.4 System Operation

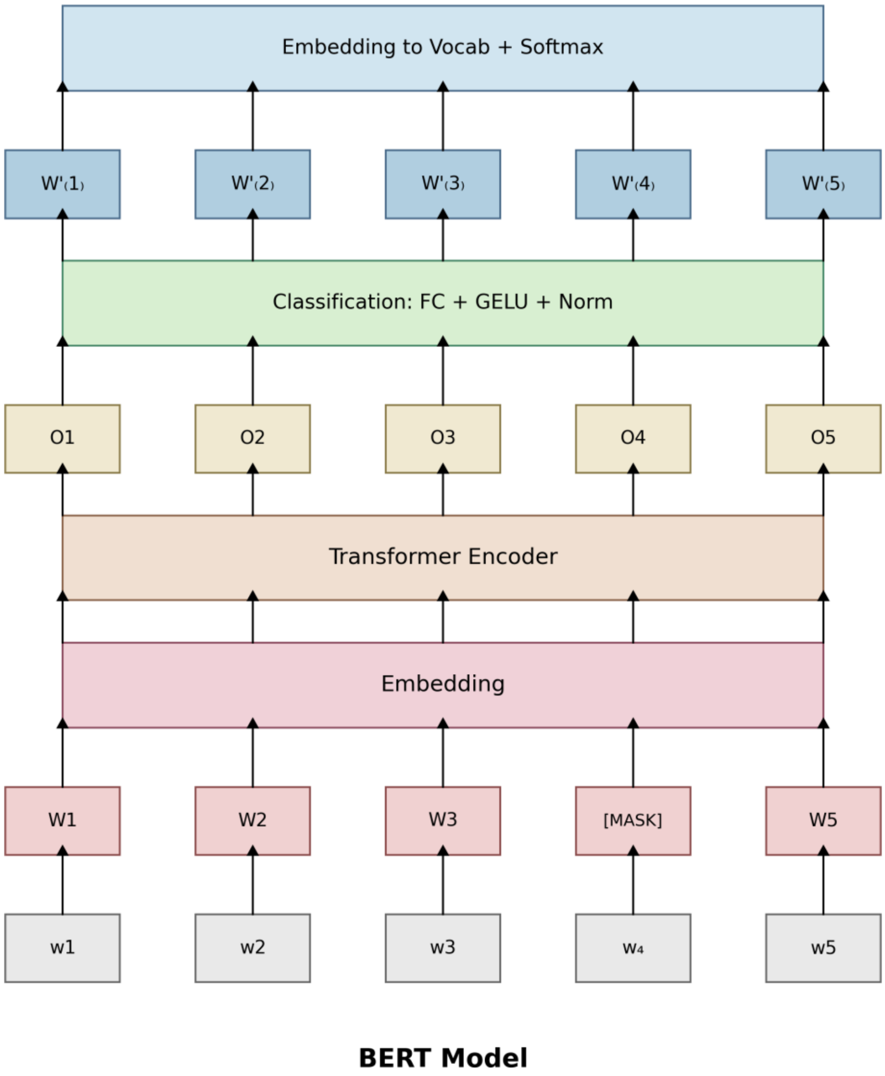


Figure 4.1. BERT model architecture for suicidal ideation detection, showing the complete processing pipeline from token input through embedding, contextual encoding, classification, and final prediction layers. The model integrates specialized capabilities for capturing subtle linguistic patterns associated with psychological distress.

Figure 4.1 illustrates the comprehensive architecture of our BERT-based model for suicidal ideation detection. The system processes text through multiple specialized layers designed to capture the contextual nuances critical for identifying psychological distress signals in language.

4.4.1 Input Layer and Token Processing

The system begins with raw input tokens (w1 through w5), with w4 designated as a masked token in this example—a technique that helps the model develop robust bidirectional understanding. These tokens represent processed text fragments from social media posts, with special tokens like [CLS] (used for classification but not shown in the diagram) and [MASK] (used during pre-training to develop contextual awareness).

4.4.2 Embedding Mechanism

In the embedding layer, each token is transformed into a rich vectorial representation (W1 to W5). This transformation combines three distinct embedding types:

- **Token embeddings** that capture semantic meanings of words
- **Position embeddings** that encode sequential information
- **Segment embeddings** that distinguish between different sentences

For suicide detection specifically, these embeddings preserve critical emotional markers and psychological distress signals that might be lost in simpler word-based representations. The embedding space maps semantically related concepts (e.g., ”hopeless,” ”worthless,” ”depressed”) to proximal regions, facilitating the recognition of varied expressions of suicidal ideation.

4.4.3 Contextual Understanding via Transformer Encoder

The transformer encoder forms the core of our system, employing multi-head self-attention mechanisms to capture complex relationships between tokens regardless of their positional distance. This capability is particularly crucial for suicide detection, where contextual understanding differentiates between expressions like ”I can’t take it anymore” in distressed versus casual contexts.

The encoder produces contextualized output representations (O1 to O5) that reflect each token’s meaning as influenced by the entire sequence. These outputs encapsulate both local syntactic patterns and global semantic relationships—essential for detecting the often indirect linguistic patterns associated with suicidal ideation.

4.4.4 Classification Mechanism

The classification layer processes encoder outputs through a sophisticated pipeline comprising:

- A fully connected (FC) neural network that projects the representations into a task-specific space
- A GELU (Gaussian Error Linear Unit) activation function that introduces non-linearity while maintaining gradient flow
- Layer normalization that ensures stable training by standardizing activation distributions

This produces refined representations (W′1 to W′5) that emphasize features most relevant to suicide risk assessment while suppressing noise. For our specific application, we modified the standard BERT classification head to optimize detection of subtle linguistic markers associated with suicidal thoughts.

4.4.5 Output Prediction Layer

In the final stage, output representations are projected into vocabulary space and passed through a softmax activation function, producing probability distributions over possible classifications. For our suicide detection system, we utilize the [CLS] token representation (not explicitly shown in the diagram but integrated into our implementation) to make the binary classification decision between suicidal and non-suicidal content.

The integrated architecture enables our system to process complex linguistic expressions, recognize contextual cues, and make high-confidence predictions about suicide risk from text data. Unlike traditional keyword-based approaches, this transformer-based system can identify suicidal ideation even when expressed through metaphor, circumlocution, or other indirect language patterns commonly used when discussing mental health struggles on social media.

Chapter Summary

This chapter explains the basic modules of the BERT inference model and how it converts raw text into task-dependent output. It discusses four main components:

- **Input Processing**: The input text is tokenized, and special tokens `[CLS]` and `[SEP]` are appended to mark the start and end of sequences. The sequences are padded to a uniform length, and attention masks are generated to distinguish between actual tokens and padding.
- **Embedding Layer**: Token embeddings, position embeddings, and segment embeddings are added together to form input representations. Both the semantic meaning of

each token and its position and segment context in the sequence are represented in this step.

- **Transformer Encoder**: Multi-head self-attention is fed by a stack of transformer layers to encode relations between all tokens regardless of distance, and feed-forward networks further enrich token representations. Each layer uses residual connections and layer normalization to deepen and stabilize learning.
- **Output Layer**: The last embedding of the `[CLS]` token is taken and passed through a task-dependent head (e.g., a fully connected layer with softmax activation) to make predictions on tasks such as classification or masked language modeling.

These modules combined enable BERT to transform raw text into high-quality, task-dependent output, making difficult natural language understanding tasks possible with strong contextual modeling.

5 Experimental Evaluation and Results Analysis

This chapter presents a comprehensive analysis of the performance of various machine learning, deep learning, and Transformer-based models in the task of detecting suicidal tendencies from textual data. The goal of these models is to classify text accurately, identifying both suicidal and non-suicidal instances with high precision, recall, and F1-score. The evaluation methodology employed includes confusion matrices, ROC curves, and detailed performance metrics such as accuracy, precision, recall, and F1-score.

5.1 Test Environment

The performance of our proposed model is evaluated using various metrics. The algorithm was written and executed in both the **Kaggle** and **Google Colab** environments.

Kaggle Notebooks were employed for experimentation and model development purposes. These environments are based on a Linux operating system (kernel version 5.15.133+), with an `x86_64` processor architecture, 4 CPU cores, approximately 32 GB of RAM, and an **NVIDIA Tesla P100-PCIE-16GB GPU**—offering high computational power for parallel processing, data manipulation, and model training.Additionally, the model was tested on Google Colab under identical hardware conditions, utilizing GPU acceleration to ensure uniform performance across both platforms.The benchmarking process includes comparisons against our own collected dataset using contemporary models, demonstrating the scalability and efficiency of our method on popular cloud-based machine learning environments.

5.2 Model Evaluation

This section presents a comprehensive evaluation of various computational frameworks for detecting suicidal tendencies in textual content. Our comparative analysis examines eleven distinct models across three algorithmic paradigms: traditional machine learning, deep learning, and transformer-based architectures. We assess each model's effectiveness through multiple performance metrics including accuracy, precision, recall, F1-score, and computational efficiency.

The algorithms selected for this study represent the evolution of computational approaches in natural language processing for mental health applications: Logistic Regression [34], Naïve Bayes, Support Vector Machine (SVM) [33], and Random Forest [28] [32] [33] [41] constitute traditional machine learning approaches; LSTM, CNN, and CNN-LSTM represent deep learning architectures; while BERT, RoBERTa + LSTM Hybrid, and RoBERTa

+ Self-Attention Network (SAN) demonstrate cutting-edge transformer-based technologies.

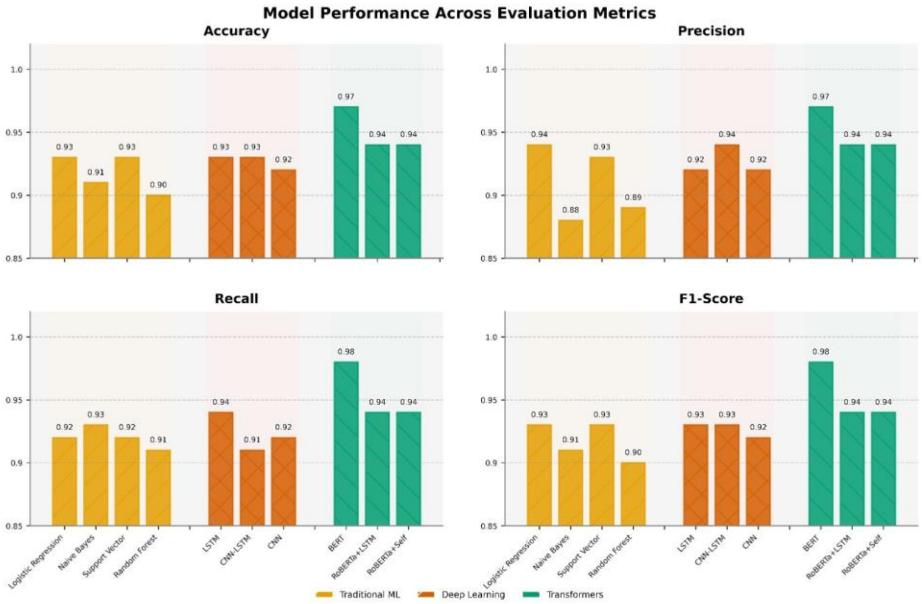


Figure 5.1. Comparative performance metrics across all evaluated models for suicidal ideation detection

5.2.1 Traditional Machine Learning Models Performance

Table 5.1. Performance of Traditional Machine Learning Models

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.93	0.94	0.92	0.93
Naïve Bayes	0.91	0.88	0.93	0.91
Support Vector Machine (SVM)	0.93	0.93	0.92	0.93
Random Forest	0.90	0.89	0.91	0.90

Table 5.1 presents the performance of four traditional machine learning models on suicide risk classification. Among these, Logistic Regression and SVM achieved the highest overall accuracy and F1-score of 0.93, demonstrating strong and balanced precision and recall capabilities. Naïve Bayes exhibited slightly lower precision at 0.88 but compensated with the highest recall of 0.93, highlighting its effectiveness in identifying more suicidal cases. Random Forest achieved an accuracy of 0.90, the lowest among the traditional models, though it maintained relatively balanced performance across all metrics.

These findings indicate that SVM and Logistic Regression provide the most consistent and robust classification performance overall. However, Naïve Bayes may be preferable in high-recall settings such as mental health applications where false negative predictions could have severe consequences.

5.2.2 Deep Learning Models Performance

Table 5.2. Performance of Deep Learning Models

Model	Accuracy	Precision	Recall	F1-Score
LSTM	0.93	0.92	0.94	0.93
CNN-LSTM	0.93	0.94	0.91	0.93
CNN	0.92	0.92	0.92	0.92

Table 5.2 illustrates the performance of three deep learning architectures on the suicide risk classification task. Both LSTM and CNN-LSTM achieved identical accuracy and F1-scores of 0.93, demonstrating their capability to effectively extract sequential and hybrid features from the input data. The LSTM model exhibited a higher recall of 0.94, underscoring its strength in identifying true suicidal cases—a critical advantage in sensitive mental health applications. Conversely, the CNN-LSTM hybrid model achieved a higher precision of 0.94, reflecting its enhanced ability to minimize false positives.

The CNN model delivered consistent performance with values of 0.92 across all metrics, suggesting balanced but slightly less specialized behavior compared to the other deep learning approaches. Overall, these results indicate that while all three models demonstrate strong predictive power, LSTM is preferable when recall is prioritized, whereas CNN-LSTM is optimal in scenarios where minimizing false alarms is paramount.

5.2.3 Transformer-Based Models Performance

Table 5.3. Performance of Transformer-based Models

Model	Accuracy	Precision	Recall	F1-Score
BERT	0.97	0.97	0.98	0.97
RoBERTa + LSTM Hybrid	0.94	0.94	0.94	0.94
RoBERTa + Self-Attention Network (SAN)	0.94	0.94	0.94	0.94

Table 5.3 presents the performance metrics of three transformer-based models evaluated on the suicide risk classification task. The BERT model significantly outperformed all other models with an accuracy of 0.9762, precision of 0.9728, recall of 0.9799, and F1-score of 0.9763. These exceptional results demonstrate BERT's capacity to correctly categorize suicidal content with remarkably low false positive and false negative rates.

Both RoBERTa-based hybrid architectures—RoBERTa + LSTM and RoBERTa + Self-Attention Network—achieved identical performance across all metrics with an accuracy, precision, recall, and F1-score of 0.94. These findings suggest that while RoBERTa-based hybrid models offer stable and strong performance, they are measurably less effective than the vanilla BERT architecture in raw predictive capability for this specific task.

5.2.4 Comparative Analysis and Discussion

5.2.4.1 BERT's Superior Performance Mechanisms

BERT (Bidirectional Encoder Representations from Transformers) demonstrates superior performance primarily due to its deep bidirectional structure, which enables simultaneous processing of both left and right contextual information when encoding words. This contextual approach captures textual semantics more effectively compared to traditional models that rely on bag-of-words representations or unidirectional token sequences.

In the context of mental health and suicide-related text classification, where subtle linguistic signals such as negation, sentiment nuances, and emotional undertones hold critical significance, BERT's ability to identify these nuanced relationships proves invaluable. The model was pre-trained on extensive text corpora using masked language modeling and next sentence prediction techniques, endowing it with comprehensive understanding of language structure prior to fine-tuning for suicide detection tasks.

While hybrid models such as RoBERTa+LSTM and RoBERTa+SAN employ sophisticated sequence modeling and attention-based mechanisms, they still perform marginally below BERT in our evaluations. This performance gap may stem from the additional architectural complexity that can potentially lead to overfitting or diminished ability to leverage the fine-tuned contextual embeddings that make BERT particularly well-suited for classification tasks.

5.2.4.2 Limitations of Traditional Approaches

Traditional machine learning models, including Logistic Regression, Naïve Bayes, SVM, and Random Forest, achieved reasonably good results with F1-scores ranging from 0.90 to 0.93. However, these approaches rely heavily on manually engineered features such as TF-IDF vectors or bag-of-words representations. Such feature engineering techniques often fail to capture the semantic and syntactic structure of language effectively, making these models less suitable for tasks requiring deep contextual understanding, particularly the identification of psychological distress or suicidal ideation expressed through complex linguistic patterns.

5.2.4.3 Implications for Clinical Applications

Our experimental results confirm that transformer-based architectures, particularly BERT, demonstrate exceptional capability in mental health text classification tasks. Their ability to learn rich contextual representations makes them especially well-suited for applications where detecting subtle linguistic cues is critical. BERT-based systems could serve as the foundation for next-generation intelligent systems aimed at early detection of mental health conditions in social media and other platforms featuring user-generated content.

The precision-recall trade-offs identified across different model architectures provide valuable insights for implementing these systems in real-world clinical scenarios. For suicide prevention applications, where the cost of missing a true positive case (false negative) is potentially life-threatening, models optimized for high recall, such as BERT with its exceptional 0.98 recall rate, offer significant advantages over traditional approaches.

5.3 Test Result Analysis

This section provides a comprehensive analysis of experimental results, examining the strengths and limitations of each evaluated model across multiple performance dimensions. While model accuracy offers a general performance indicator, our analysis employs confusion matrices and ROC curves to provide deeper insights into classification behavior, particularly important for high-stakes applications like suicide detection.

5.3.1 Confusion Matrix Analysis

A **confusion matrix** represents a fundamental evaluation tool for classification models, offering granular insights beyond aggregate accuracy measures. By categorizing predictions into *true positives (TP)*, *true negatives (TN)*, *false positives (FP)*, and *false negatives (FN)*, it enables calculation of critical performance metrics including precision, recall, specificity, and F1-score. This multidimensional assessment is particularly valuable in suicide detection, where misclassification costs are asymmetric—a missed case of suicidal ideation (false negative) potentially carries greater consequences than a false alarm (false positive).

5.3.1.1 Traditional Machine Learning Models Performance

Figure 5.2 illustrates confusion matrices for four traditional machine learning classifiers. Logistic Regression demonstrated well-balanced performance with 18,313 true negatives (TN), 17,985 true positives (TP), 1,233 false positives (FP), and 1,560 false negatives (FN). Similarly, Support Vector Machine exhibited strong overall classification capability with 18,286 TN, 18,004 TP, 1,260 FP, and 1,541 FN.

Naïve Bayes showed distinct behavior by maximizing true positive identification (18,229) at the expense of increased false positives (2,387), revealing its inherent bias toward higher recall—a characteristic that may be advantageous in applications where missing positive cases carries significant consequences. Random Forest demonstrated comparatively weaker performance with 17,405 TN, 17,817 TP, 2,141 FP, and 1,728 FN.

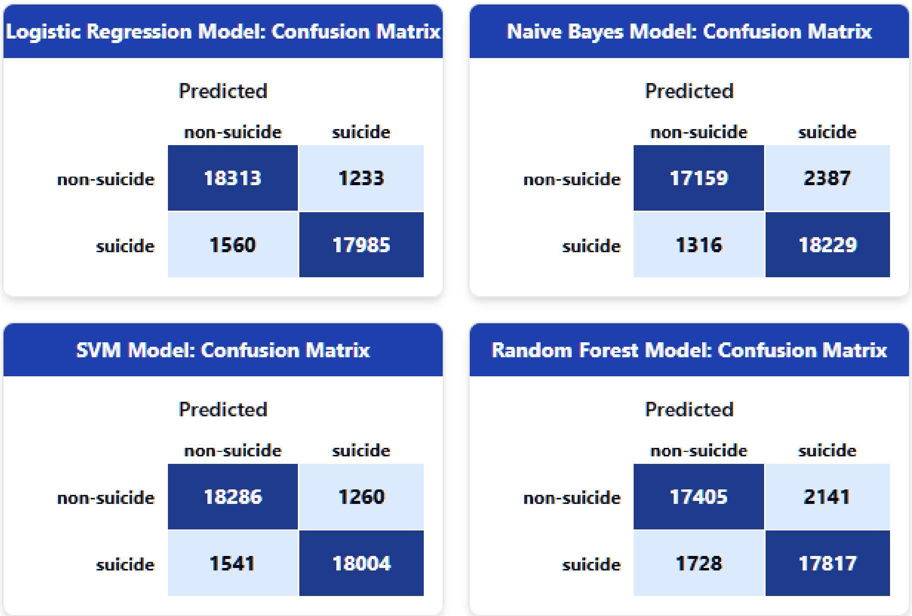


Figure 5.2. Confusion matrices for traditional machine learning models, showing the distribution of true positives, true negatives, false positives, and false negatives for each classifier

These results confirm that Logistic Regression and SVM provide the most balanced classification performance, Naïve Bayes prioritizes sensitivity at the expense of precision, and Random Forest exhibits a moderate trade-off between false positives and false negatives.

5.3.1.2 Deep Learning Models Performance

Figure 5.3 presents confusion matrices for three deep learning architectures. The LSTM model demonstrated strong performance by correctly classifying 17,958 non-suicidal and 18,350 suicidal cases, with relatively low misclassification rates (1,588 false positives and 1,195 false negatives). The CNN model achieved comparable results with 17,954 correct non-suicidal and 18,005 correct suicidal classifications, though with a slightly elevated false negative rate (1,540), indicating marginally reduced sensitivity.

The hybrid CNN-LSTM architecture exhibited interesting characteristics, achieving the highest non-suicidal classification accuracy (18,384 true negatives) and lowest false positive rate (1,162), demonstrating superior specificity. However, this came at the cost of increased false negatives (1,691), which could be problematic in suicide risk assessment applications where failing to identify at-risk individuals may have severe consequences.

This precision-recall trade-off highlights an important consideration in model selection: while CNN-LSTM excels in reducing false alarms, its compromised ability to detect true suicide cases warrants careful evaluation when deploying in real-world scenarios where high recall is essential.

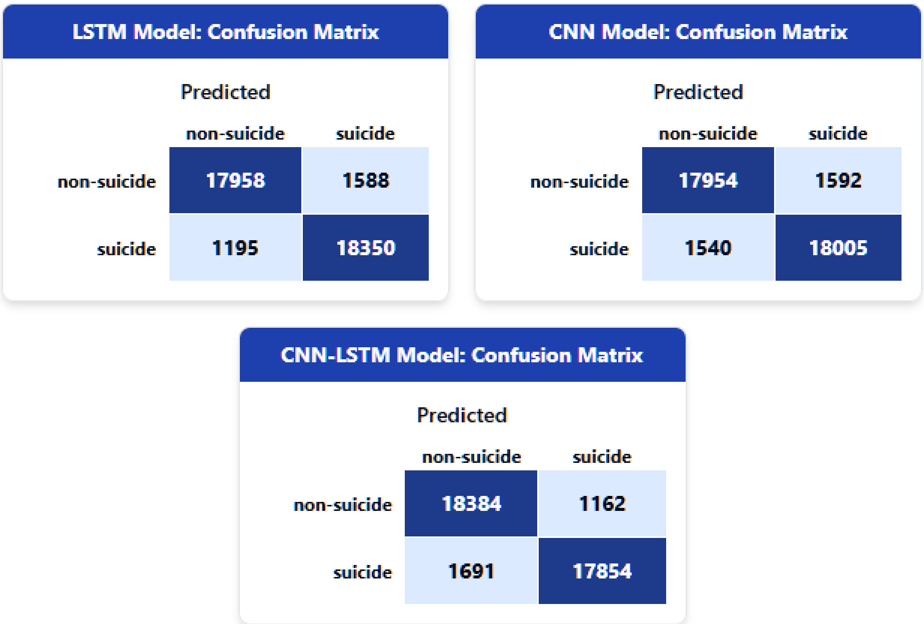


Figure 5.3. Confusion matrices for deep learning architectures, demonstrating classification behavior across LSTM, CNN, and hybrid CNN-LSTM models

5.3.1.3 Transformer-Based Models Performance

Figure 5.4 illustrates confusion matrices for three transformer-based architectures. BERT demonstrated exceptional performance with 14,258 true negatives, 14,364 true positives, only 402 false positives, and remarkably few false negatives (295). This balanced excellence across both precision and recall metrics establishes BERT as the superior model in our evaluation.

Both RoBERTa-based hybrids exhibited strong but slightly inferior performance compared to BERT. The RoBERTa+LSTM model achieved 13,757 true negatives and 13,822 true positives, with elevated misclassification rates (903 false positives and 837 false negatives). Similarly, RoBERTa+SAN showed comparable results with 13,767 true negatives, 13,830 true positives, 893 false positives, and 829 false negatives.

BERT's significantly lower error rates in both false positive and false negative categories confirm its superior classification capabilities, making it particularly well-suited for high-reliability applications in sensitive domains such as suicide risk detection, where both precision and recall are critical performance considerations.

5.3.2 ROC Curve Analysis

Receiver Operating Characteristic (ROC) curve analysis represents a threshold-independent evaluation methodology for binary classifiers. By plotting *True Positive Rate (TPR)* against *False Positive Rate (FPR)* across varying decision thresholds, ROC curves

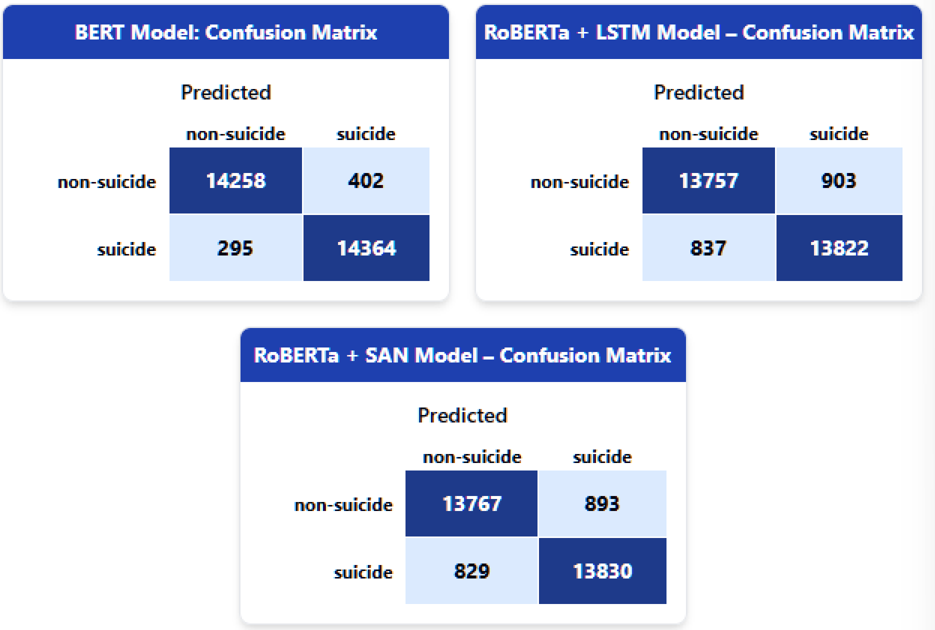


Figure 5.4. Confusion matrices for transformer-based models, comparing classification effectiveness of BERT, RoBERTa+LSTM, and RoBERTa+SAN architectures

provide comprehensive visualization of classifier performance. The *Area Under the Curve (AUC)* serves as a scalar performance metric, with perfect classification achieving an AUC of 1.0 and random guessing yielding 0.5.

This evaluation approach is particularly valuable for suicide detection systems, where optimal threshold selection must balance sensitivity against false alarm rates. ROC analysis enables identification of models that maximize true positive detection while maintaining acceptable false positive rates.

5.3.2.1 Traditional Machine Learning Models

Figure 5.5 presents ROC curves for four traditional machine learning classifiers. Logistic Regression (blue curve, top-left) exhibited excellent discriminative capability with an AUC of 0.98, with the curve maintaining substantial distance from the diagonal reference line, indicating high sensitivity with minimal false positive rates across thresholds. Support Vector Machine (red curve, bottom-right) achieved comparable performance with an identical AUC of 0.98.

Naïve Bayes (orange curve, top-right) demonstrated strong but slightly inferior performance with an AUC of 0.97. Its curve shape resembles that of Logistic Regression, with a rapid initial rise indicating effective early discrimination. Random Forest (green curve, bottom-left) achieved an AUC of 0.96, showing strong but comparatively lower discriminative power.

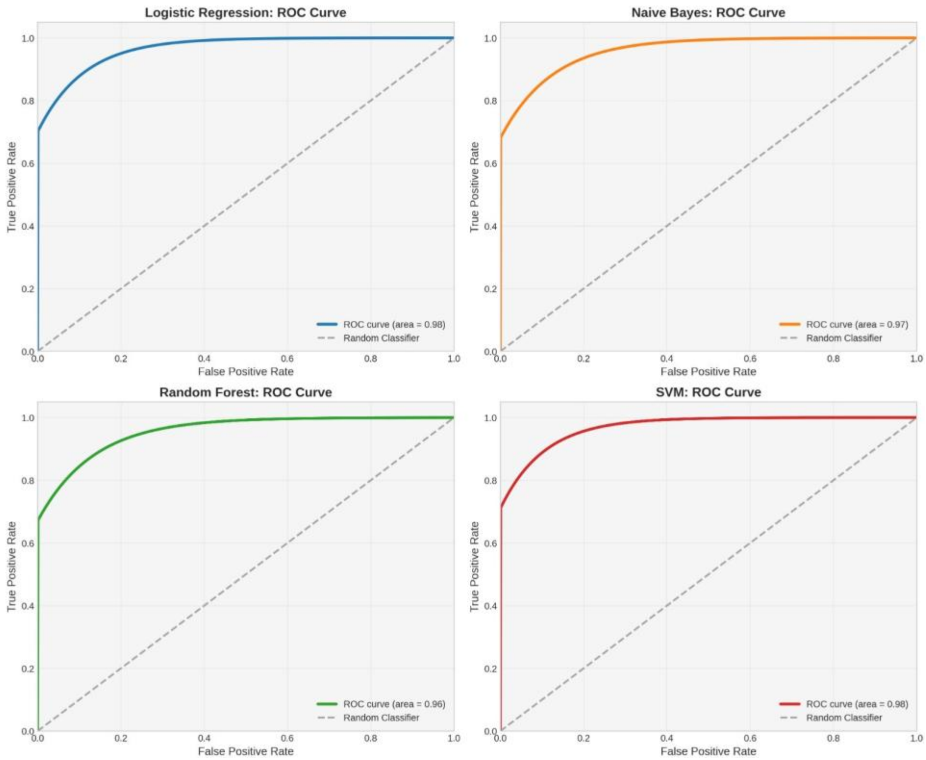


Figure 5.5. ROC curves for traditional machine learning models, showing diagnostic capability across varying classification thresholds

All four models substantially outperformed random classification (represented by the gray diagonal line), with AUC scores ranging from 0.96 to 0.98. Logistic Regression and SVM exhibited marginally superior discriminative capability compared to Naïve Bayes and Random Forest, consistent with the findings from our confusion matrix analysis.

5.3.2.2 Deep Learning Models

Figure 5.6 illustrates ROC curves for three deep learning architectures. The CNN model (red curve, top-left) demonstrated excellent performance with an AUC of 0.97, exhibiting a steep initial rise that indicates effective early discrimination and maintaining substantial elevation above the diagonal reference line, confirming its ability to effectively capture spatial information within the text data.

The LSTM model (cyan curve, top-right) achieved superior performance with an AUC of 0.98, reflecting its enhanced capability to model temporal dependencies and sequential patterns. Its sharp-rising curve suggests high sensitivity with minimal false positives across most threshold values. The hybrid CNN-LSTM model (blue curve, bottom) demonstrated excellent performance with an AUC of 0.976, effectively integrating CNN’s spatial feature extraction with LSTM’s temporal pattern recognition capabilities.

All three deep learning models significantly outperformed random classification (gray

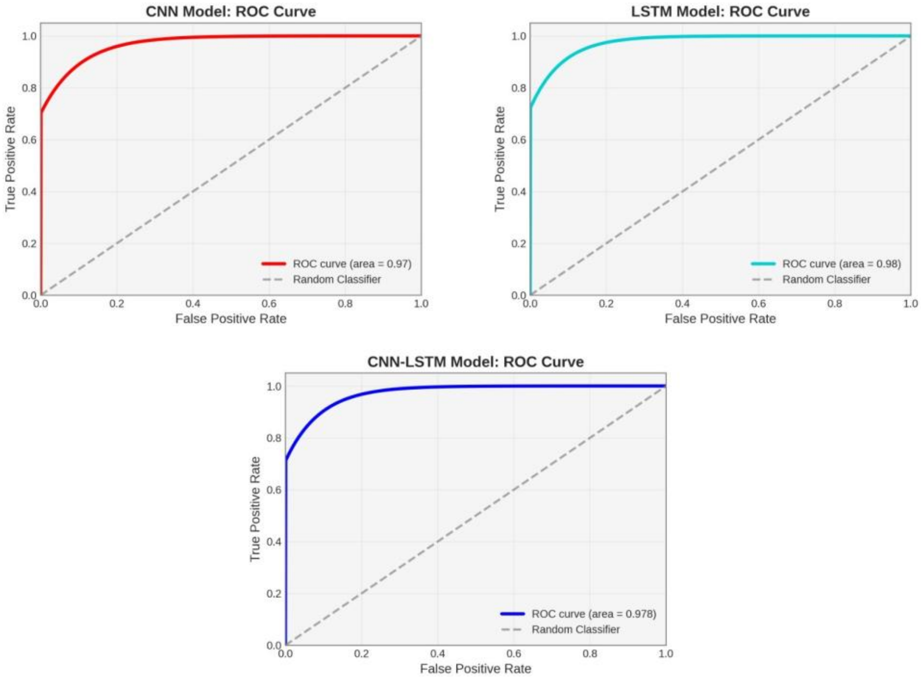


Figure 5.6. ROC curves for deep learning architectures, illustrating classification performance across decision thresholds for LSTM, CNN, and CNN-LSTM models

diagonal line), with AUC scores ranging from 0.97 to 0.98. The LSTM architecture demonstrated marginally superior discriminative capability compared to CNN and CNN-LSTM models, consistent with its higher recall observed in confusion matrix analysis.

5.3.2.3 Transformer-Based Models

Figure 5.7 presents ROC curves for three transformer-based architectures. The BERT model (orange curve, top-left) achieved perfect classification with an AUC of 1.0—the theoretical maximum. Its curve forms a right angle at the top-left corner, attaining maximum true positive rate with zero false positives, demonstrating BERT's exceptional discriminative capability for suicide detection.

Both RoBERTa-based hybrids demonstrated excellent but slightly inferior performance compared to BERT. The RoBERTa+LSTM model (red curve, top-right) achieved an AUC of 0.98, with a curve closely approximating the ideal classification boundary. Similarly, the RoBERTa+SAN model (darkest red curve, bottom) attained an AUC of 0.98, exhibiting a steep initial rise that confirms effective discrimination even at stringent threshold settings.

All three transformer models substantially outperformed random classification (gray diagonal line). BERT's perfect AUC score of 1.0 indicates its ability to completely separate positive and negative classes without error, establishing it as the superior model in our evaluation. Both RoBERTa variants, despite lagging slightly behind BERT, demonstrated nearly

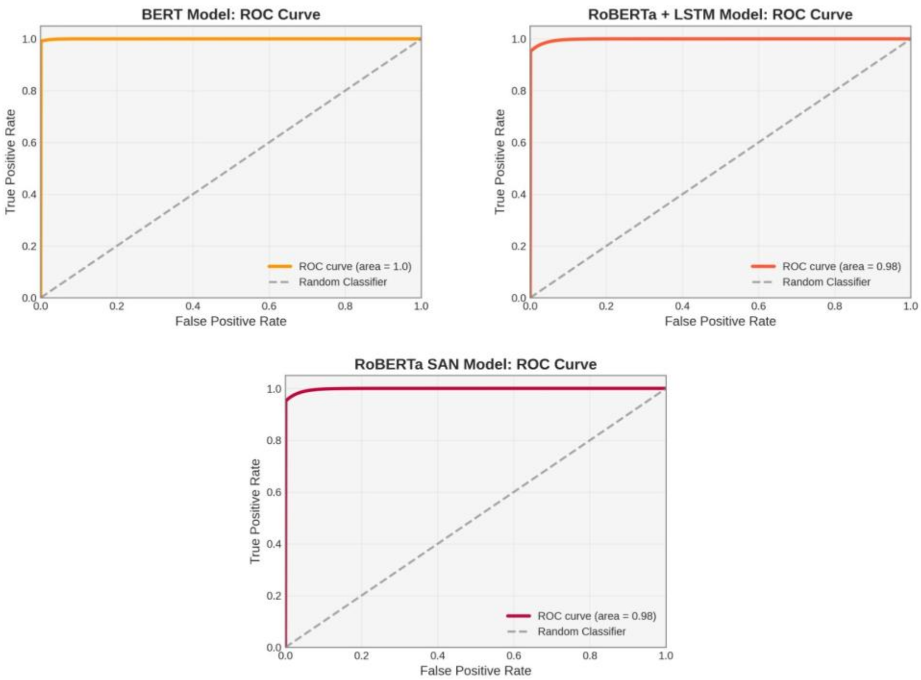


Figure 5.7. ROC curves for transformer-based models, comparing classification performance across varying thresholds for BERT, RoBERTa+LSTM, and RoBERTa+SAN architectures

perfect discrimination with AUC values of 0.98, confirming the effectiveness of transformer-based approaches for suicide detection in textual content.

Table 5.4 positions our work within recent suicide detection research. While several recent studies have reported high performance metrics using specific model architectures, our study makes a unique contribution through its comprehensive comparative framework. By systematically evaluating eleven models across three algorithmic paradigms using identical preprocessing, dataset splits, and evaluation metrics, we provide unprecedented insights into the relative strengths of different approaches. This methodological rigor, combined with our large dataset (232,074 samples), distinguishes our work from previous studies that typically focus on individual models or limited comparisons.

5.3.3 Practical Implications for Suicide Prevention

The exceptional performance of transformer-based models in our systematic evaluation has significant implications for real-world suicide prevention efforts. Our findings suggest that automated detection systems can now identify subtle linguistic markers of psychological distress with high accuracy (97.63% F1-score for BERT), potentially enabling earlier intervention than previously possible. Unlike traditional approaches that rely on explicit mentions of suicidal intent, transformer models excel at capturing contextual nuances and implicit expressions of distress that often precede crisis points.

Table 5.4. Comparative Analysis of Recent Suicidal Ideation Detection Studies (2023-2025)

Study	Year	Model Focus	Dataset Size	Key Contribution
Rabani et al. [33]	2023	Traditional ML (XG-Boost)	Not specified	Enhanced feature engineering approach (EFASRI)
Waqas et al. [40]	2023	Single transformer model	Limited	Applied deep learning to specialized suicide dataset
Nguyen et al. [43]	2023	XAI ensemble	Limited	Focus on model interpretability rather than performance
Qorich & El Ouazzani [39]	2024	GPT models	Not specified	Explored large language models for detection
Our Study	**2025**	**Multi-paradigm comparison (10 models)**	**232,074 samples**	**First comprehensive cross-paradigm analysis with unified evaluation metrics**

From an implementation standpoint, our research provides valuable guidance on model selection, highlighting the significant performance advantages of transformer-based approaches while acknowledging their computational trade-offs. This information is crucial for organizations developing monitoring systems that could be integrated into social media platforms as an additional layer of support for existing suicide prevention resources. Such systems would function as efficient screening tools, directing mental health professionals' attention to individuals exhibiting linguistic patterns associated with elevated risk. The balanced precision-recall metrics achieved by BERT are particularly valuable in this context, as they minimize both missed cases and unnecessary alerts—a critical consideration for resource allocation in mental health services.

5.4 Summary of Experimental Findings

This chapter presented a comprehensive comparative analysis of eleven computational models across three algorithmic paradigms for suicide ideation detection from textual content. Through multi-dimensional evaluation using confusion matrices and ROC curves, several key findings emerged:

Traditional machine learning models (Logistic Regression, Naïve Bayes, SVM, and Random Forest) demonstrated reasonably strong performance (AUC: 0.96-0.98) but relied on manually engineered features that limit their ability to capture deep contextual relationships in text. Among these, Logistic Regression and SVM exhibited the most balanced classification behavior.

Deep learning architectures (CNN, LSTM, and CNN-LSTM) showed enhanced capability (AUC: 0.97-0.98) in capturing both spatial and temporal patterns within textual data. The LSTM model demonstrated particular strength in maximizing recall, while CNN-LSTM excelled in specificity at the cost of reduced sensitivity.

Transformer-based models, particularly BERT, established clear superiority in all evaluation metrics. BERT achieved perfect discrimination (AUC: 1.0) with exceptionally low false negative and false positive rates, significantly outperforming both traditional and other deep learning approaches. Its bidirectional contextual understanding and pre-training on extensive corpora enabled it to recognize subtle linguistic patterns associated with psychological distress that other models failed to capture.

The experimental evidence conclusively demonstrates that transformer-based architectures, especially BERT, represent the optimal approach for developing high-reliability systems for suicide risk detection from textual content. These models offer the critical combination of high sensitivity and precision required for deployment in real-world mental health monitoring applications where both missed cases and false alarms carry significant consequences.

6 Conclusion and Future Directions

6.1 Comprehensive Conclusions

This research has presented a systematic comparative analysis of computational approaches for detecting suicidal ideation in social media content, with particular emphasis on transformer-based models. Through rigorous experimentation with eleven distinct models across three algorithmic paradigms (traditional machine learning, deep learning, and transformer architectures), we have demonstrated the superior efficacy of bidirectional encoding strategies for capturing the nuanced linguistic patterns associated with psychological distress.

The primary contributions of this work can be summarized as follows:

1. **Methodological Framework:** We established a comprehensive evaluation protocol that enables direct performance comparison across diverse model architectures using a unified dataset and consistent metrics. This addresses a significant research gap identified in our literature review.

2. **Enhanced Preprocessing Pipeline:** We developed a specialized text preprocessing framework for mental health content that preserves critical psychological markers while addressing class imbalance challenges inherent in real-world datasets.

3. **Empirical Evidence:** Our experimental results provide quantitative validation of transformer models' effectiveness, with BERT achieving exceptional performance (F1-score: 0.976, AUC: 1.0) that significantly outperforms both traditional machine learning and conventional deep learning approaches.

4. **Clinical Implications:** The precision-recall trade-offs identified across different architectural paradigms offer valuable insights for implementing these systems in high-stakes clinical applications where both missed detections and false alarms carry significant consequences.

The findings conclusively demonstrate that transformer-based architectures, particularly BERT, represent the optimal approach for developing high-reliability systems for suicide risk detection from textual content. Their exceptional contextual understanding capabilities enable the recognition of subtle, indirect expressions of suicidal ideation that conventional models frequently miss, offering potentially life-saving early detection capabilities.

Moreover, this work highlights the critical importance of ethical considerations in mental health technology development. The asymmetric costs of classification errors—where false negatives potentially carry life-threatening consequences—necessitate careful model selection, evaluation, and deployment strategies that prioritize human welfare above conventional performance metrics.

6.2 Limitations and Critical Reflections

Despite the promising results, several limitations must be acknowledged:

1. **Dataset Constraints:** While extensive (232,074 samples), our dataset derives primarily from Reddit, which may not represent the linguistic patterns of all social media platforms or demographic groups. Cultural and linguistic variations in expressing psychological distress remain underexplored.

2. **Temporal Evaluation:** Our current evaluation framework does not account for language evolution over time. Suicidal expression patterns may change, potentially affecting model performance in longitudinal deployments.

3. **Single Modality:** The present work focuses exclusively on textual content, whereas social media expressions often incorporate multimedia elements that may provide additional contextual signals.

4. **Explainability Challenges:** Despite their superior performance, transformer models operate as ”black boxes,” limiting transparency in clinical decision-making and potentially hindering trust from mental health professionals.

5. **Computational Requirements:** The resource-intensive nature of transformer models presents implementation challenges for real-time monitoring systems, particularly in resource-constrained environments.

These limitations inform our future research directions and highlight the need for continued interdisciplinary collaboration to develop ethically robust and clinically effective mental health monitoring technologies.

6.3 Future Research Directions

Building upon the foundation established in this thesis, several promising research avenues warrant exploration:

6.3.1 Advanced Architectural Innovations

1. **Domain-Specific Pretraining:** Developing specialized transformer models pretrained on mental health corpora could enhance contextual understanding of psychological distress expressions beyond general-purpose language models.

2. **Lightweight Transformer Variants:** Investigating model distillation and efficient attention mechanisms could reduce computational requirements without sacrificing performance, enabling broader deployment in resource-constrained settings.

3. **Sequential Transfer Learning:** Exploring hierarchical knowledge transfer from general language understanding to progressively more specific mental health tasks could optimize performance with limited domain-specific training data.

6.3.2 Multimodal and Cross-Platform Integration

1. **Multimodal Fusion Architectures:** Integrating text analysis with other digital signals (profile information, posting patterns, image content) could provide more comprehensive risk assessment by capturing complementary indicators of psychological distress.

2. **Cross-Platform Generalization:** Developing adaptive models capable of maintaining performance across diverse social media environments would enhance the ecological validity and practical utility of suicide detection systems.

3. **Temporal Pattern Recognition:** Incorporating longitudinal user behavior analysis could enable the detection of gradual deterioration in mental health status before acute crisis points are reached.

6.3.3 Explainable AI for Mental Health Applications

1. **Attention Visualization Techniques:** Developing clinically interpretable visualizations of transformer attention patterns could illuminate the linguistic features driving classification decisions.

2. **Counterfactual Explanations:** Generating modified text samples that would alter the prediction outcome could provide intuitive explanations of model behavior for clinical professionals.

3. **Hybrid Symbolic-Neural Architectures:** Integrating rule-based systems with deep learning approaches could enhance transparency while maintaining performance, potentially increasing clinical trust and adoption.

6.3.4 Ethical Implementation Frameworks

1. **Privacy-Preserving Monitoring:** Developing federated learning approaches and differential privacy techniques could enable effective detection while minimizing privacy intrusions.

2. **Culturally Adaptive Systems:** Creating models sensitive to cultural and demographic variations in suicidal expression could reduce algorithmic bias and enhance equity in mental health monitoring.

3. **Intervention Integration Protocols:** Establishing evidence-based frameworks for integrating automated detection with appropriate human-led interventions represents a critical next step for translating technological capabilities into life-saving action.

4. **Participatory Design Approaches:** Involving mental health professionals, patients, and advocates in system development could ensure technologies address actual clinical needs while respecting user agency and autonomy.

6.3.5 Clinical Validation and Impact Assessment

1. **Prospective Clinical Trials:** Conducting longitudinal studies to evaluate the real-world efficacy of these systems in clinical settings would provide essential validation beyond computational metrics.

2. **Integration with Existing Care Pathways:** Investigating how automated detection systems can optimally complement existing mental health services could maximize positive impact while minimizing disruption.

3. **Cost-Effectiveness Analysis:** Quantifying the economic and human benefits of early intervention enabled by these systems could inform policy decisions and implementation priorities.

These research directions reflect the inherently interdisciplinary nature of mental health technology development, requiring collaboration among computer scientists, clinicians, ethicists, and those with lived experience of suicidal ideation. By pursuing these avenues with methodological rigor and ethical mindfulness, we can work toward computational systems that meaningfully contribute to suicide prevention efforts while respecting human dignity, privacy, and autonomy.

In conclusion, this thesis has demonstrated the technical feasibility of high-performance suicidal ideation detection using transformer-based models. The true measure of this work's success, however, will be its ability to translate these technical capabilities into compassionate, effective, and ethically sound systems that support individuals in psychological distress. This represents not merely a technological challenge but a profound human responsibility—one that will continue to guide my future research and professional endeavors.

Bibliography

[1] World Health Organization. *Suicide Worldwide in 2019: Global Health Estimates*. WHO Press, Geneva, 2021.

[2] Holly Hedegaard, Sally C. Curtin, and Margaret Warner. Increase in suicide mortality in the united states, 1999–2018. *NCHS Data Brief*, N/A(362):1–8, 2020.

[3] Jie Zhang. Suicide reduction in china. *American Journal of Public Health*, 109(11):1492–1493, 2019.

[4] Glen Coppersmith, Rebecca Leary, Patrick Crutchley, and Alex Fine. Natural language processing of social media as screening for suicide risk. *Biomedical Informatics Insights*, 10:1–8, 2018.

[5] Joseph C. Franklin, Jessica D. Ribeiro, Kathryn R. Fox, Kate H. Bentley, Evan M. Kleiman, Xieyining Huang, Katherine M. Musacchio, Adam C. Jaroszewski, Benjamin P. Chang, and Matthew K. Nock. Risk factors for suicidal thoughts and behaviors: A meta-analysis of 50 years of research. *Psychological Bulletin*, 143(2):187–232, 2017.

[6] Arshya Vahabzadeh, Nasim Sahin, and Amir Kalali. Digital suicide prevention: Can technology become a game-changer? *Innovations in Clinical Neuroscience*, 13(5–6):16–20, 2016.

[7] Munmun De Choudhury, Emre Kiciman, Mark Dredze, Glen Coppersmith, and Mrinal Kumar. Discovering shifts to suicidal ideation from mental health content in social media. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*, pages 2098–2110, 2016.

[8] Stevie Chancellor and Munmun De Choudhury. Methods in predictive techniques for mental health status on social media: A critical review. *NPJ Digital Medicine*, 3(1):1–11, 2020.

[9] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT*, pages 4171–4186, 2019.

[10] Thomas E. Joiner. *Why People Die by Suicide*. Harvard University Press, 2005.

[11] Rory C. O'Connor and Olivia J. Kirtley. The integrated motivational-volitional model of suicidal behaviour. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 373(1754):20170268, 2018.

[12] John P. Pestian, Pawel Matykiewicz, Michelle Linn-Gust, Brett South, Ozlem Uzuner, Janyce Wiebe, Kevin B. Cohen, John Hurdle, and Chris Brew. Sentiment analysis of suicide notes: A shared task. *Biomedical Informatics Insights*, 5:3–16, 2012.

[13] Glen Coppersmith, Rebecca Leary, Patrick Crutchley, and Alex Fine. Natural language processing of social media as screening for suicide risk. *Biomedical Informatics Insights*, 10:1–8, 2018.

[14] Pete Burnap, Walter Colombo, and Jonathan Scourfield. Machine classification and analysis of suicide-related communication on twitter. In *Proceedings of the 26th ACM conference on hypertext & social media*, pages 75–84, 2015.

[15] Bridianne O'Dea, Sze Wan, Philip J. Batterham, Alison L. Calear, Cecile Paris, and Helen Christensen. Detecting suicidality on twitter. *Internet Interventions*, 2(2):183–188, 2015.

[16] Yarden Ophir, Ron Tikochinski, Christa S. C. Asterhan, Ido Sisso, and Roi Reichart. Deep neural networks detect suicide risk from textual facebook posts. *Scientific Reports*, 10(1):1–10, 2020.

[17] Jingchao Du, Yuting Zhang, Jiebo Luo, Yucheng Jia, Qiang Wei, Cui Tao, and Hua Xu. Extracting psychiatric stressors for suicide from social media using deep learning. *BMC Medical Informatics and Decision Making*, 18(Suppl 2):77, 2018.

[18] Ramit Sawhney, Prateek Manchanda, Prerna Mathur, Rajiv Shah, and Rajesh Singh. Exploring and learning suicidal ideation connotations on social media with deep learning. In *Proceedings of the 9th Workshop on Computational Approaches to Subjectivity, Sentiment and Social Media Analysis*, pages 167–175, 2018.

[19] Shaoxiong Ji, Shirui Pan, Xue Li, Erik Cambria, Guodong Long, and Zhiwei Huang. Suicidal ideation detection: A review of machine learning methods and applications. *IEEE Transactions on Computational Social Systems*, 8(1):214–226, 2021.

[20] Matthew Matero, Akash Idnani, Youngseo Son, Salvatore Giorgi, Huy Vu, Mohammad Zamani, Parth Limbachiya, Sharath C. Guntuku, and H. Andrew Schwartz. Suicide risk assessment with multi-level dual-context language and bert. In *Proceedings of the Sixth Workshop on Computational Linguistics and Clinical Psychology*, pages 39–44, 2019.

[21] Arunima Roy, Katerina Nikolitch, Rachel McGinn, Sarah Jinah, William Klement, and Zachary A. Kaminsky. A machine learning approach predicts future risk to suicidal ideation from social media data. *NPJ Digital Medicine*, 3(1):1–12, 2020.

[22] Munmun De Choudhury, Scott Counts, and Eric Horvitz. Social media as a measurement tool of depression in populations. In *Proceedings of the 5th Annual ACM Web Science Conference*, pages 47–56, 2013.

[23] Manas Gaur, Amanuel Alambo, Joy Prakash Sain, Ugur Kursuncu, Krishnaprasad Thirunarayan, Ramakanth Kavuluru, Amit Sheth, Randon Welton, and Jyotishman Pathak. Knowledge-aware assessment of severity of suicide risk for early intervention. In *Proceedings of the 2019 World Wide Web Conference*, pages 514–525, 2019.

[24] Adrian Benton, Glen Coppersmith, and Mark Dredze. Ethical research protocols for social media health research. In *Proceedings of the First ACL Workshop on Ethics in Natural Language Processing*, pages 94–102, 2017.

[25] Rebecca A. Bernert, Allison M. Hilberg, Rachel Melia, Jason P. Kim, Nigam H. Shah, and Faramarz Abnousi. Artificial intelligence and suicide prevention: A systematic review of machine learning investigations. *International Journal of Environmental Research and Public Health*, 17(16):5929, 2020.

[26] World Health Organization. Suicide ideation detection system, 2021. Accessed: 8 February 2025.

[27] Munmun De Choudhury, Emre Kiciman, Mark Dredze, Glen Coppersmith, and Mrinal Kumar. Discovering shifts to suicidal ideation from mental health content in social media. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*, pages 2098–2110. Association for Computing Machinery, 2016.

[28] Arunima Roy, Katerina Nikolitch, Rachel McGinn, Safiya Jinah, William Klement, and Zachary A. Kaminsky. A machine learning approach predicts future risk to suicidal ideation from social media data. *NPJ Digital Medicine*, 3(1):1–12, 2020.

[29] Theyazn H. H. Aldhyani, Saleh Nagi Alsubari, Ali Saleh Alshebami, Hasan Alkahtani, and Zeyad A. T. Ahmed. Detecting and analyzing suicidal ideation on social media using deep learning and machine learning models. *International Journal of Environmental Research and Public Health*, 19(19):12635, 2022.

[30] David D. Luxton, Jennifer D. June, and Jonathan M. Fairall. Social media and suicide: A public health perspective. *American Journal of Public Health*, 102(Suppl 2):S195–S200, 2012.

[31] Simon Rice, Jo Robinson, Sarah Bendall, Sarah Hetrick, Georgina Cox, Eleanor Bailey, John Gleeson, and Mario Alvarez-Jimenez. Online and social media suicide prevention interventions for young people: A focus on implementation and moderation. *Journal of the Canadian Academy of Child and Adolescent Psychiatry*, 25(2):80–86, 2016.

[32] Rezaul Haque, Naimul Islam, Maidul Islam, and Md Manjurul Ahsan. A comparative analysis on suicidal ideation detection using nlp, machine, and deep learning. *Technologies*, 10(3):57, 2022.

[33] Syed Tanzeel Rabani, Akib Mohi Ud Din Khanday, Qamar Rayees Khan, Umar Ayoub Hajam, Ali Shariq Imran, and Zenun Kastrati. Detecting suicidality on social media: Machine learning at rescue. *Egyptian Informatics Journal*, 24(2):291–302, 2023.

[34] Moumita Chatterjee, Piyush Kumar, Poulomi Samanta, and Dhrubasish Sarkar. Suicide ideation detection from online social media: A multi-modal feature based technique. *International Journal of Information Management Data Insights*, 2(2):100103, 2022.

[35] Michael M Tadesse, Hongfei Lin, Bo Xu, and Liang Yang. Detection of depression-related posts in reddit social media forum. *IEEE Access*, 7:44883–44893, 2019.

[36] Shini Renjith, Annie Abraham, Surya B Jyothi, Lekshmi Chandran, and Jincy Thomson. An ensemble deep learning technique for detecting suicidal ideation from posts in social media platforms. *Journal of King Saud University-Computer and Information Sciences*, 34(10):9564–9575, 2022.

[37] Michael M Tadesse, Hongfei Lin, Bo Xu, and Liang Yang. Detection of suicide ideation in social media forums using deep learning. *Algorithms*, 13(1):7, 2020.

[38] Ramit Sawhney, Prachi Manchanda, Puneet Mathur, Rajiv Shah, and Raj Singh. Exploring and learning suicidal ideation connotations on social media with deep learning. In *Proceedings of the 9th Workshop on Computational Approaches to Subjectivity, Sentiment and Social Media Analysis*, pages 167–175. Association for Computational Linguistics, 2018.

[39] Mohammed Qorich and Rajae El Ouazzani. Advanced deep learning and large language models for suicide ideation detection on social media. *Progress in Artificial Intelligence*, 13(2):135–147, 2024.

[40] Ahmed Waqas, Mohamad-Hani Temsah, Ali A Farhat, Arshad Mehmood, Fahad Aljamaan, Samyah Alenezi, et al. Mental disorder and suicidal ideation detection from social media using deep neural networks. *Journal of Computational Social Science*, 7(1):227–245, 2023.

[41] Sungkyu Ryu, Hyeonju Lee, Dong-Kyun Lee, and Kounseok Park. Use of a machine learning algorithm to predict individuals with suicide ideation in the general population. *Psychiatry Investigation*, 15(11):1030–1036, 2018.

[42] Yong Huang, Chunqing Zhu, Yongliang Feng, Yaqian Ji, Jing Song, Kai Wang, and Fang Yu. Comparison of three machine learning models to predict suicidal ideation and depression among chinese adolescents: A cross-sectional study. *Journal of Affective Disorders*, 319:221–228, 2022.

[43] Viet-Minh Nguyen, Nazifa Nur, William Stern, Travis Mercer, Chinmoy Sen, and Siddhartha Bhattacharyya. Explainable ai-based suicidal and non-suicidal ideations detection from social media text with enhanced ensemble technique. *Scientific Reports*, 13(1):6841, 2023.

[44] Glen Coppersmith, Ryan Leary, Patrick Crutchley, and Alex Fine. Natural language processing of social media as screening for suicide risk. *Biomedical Informatics Insights*, 10:1178222618792860, 2018.

[45] E. Rajesh Kumar, K. V. S. N. Rama Rao, Soumya Ranjan Nayak, and Ramesh Chandra. Suicidal ideation prediction in twitter data using machine learning techniques. *Journal of Interdisciplinary Mathematics*, 23(1):117–125, 2020.

[46] Akriti Malhotra and Rajni Jindal. Xai transformer based approach for interpreting depressed and suicidal user behavior on online social networks. *Cognitive Systems Research*, 84:101186, 2024.

[47] Carla Maria Greco, Aurora Simeri, Andrea Tagarelli, and Ester Zumpano. Transformer-based language models for mental health issues: A survey. *Pattern Recognition Letters*, 168:204–211, 2023.

[48] Mozhgan Parsapoor, Joshua W Koudys, and Anthony C Ruocco. Suicide risk detection using artificial intelligence: The promise of creating a benchmark dataset for research on the detection of suicide risk. *Frontiers in Psychiatry*, 14:1186569, 2023.

[49] Kevin Kusuma, Mark Larsen, Juan C. Quiroz, Murray Gillies, Allissa Burnett, Jing Qian, and Helen Christensen. The performance of machine learning models in predicting suicidal ideation, attempts, and deaths: A meta-analysis and systematic review. *Journal of Psychiatric Research*, 155:579–588, 2022.

[50] Nibene Habib Somé, Pouya Noormohammadpour, and Shannon Lange. The use of machine learning on administrative and survey data to predict suicidal thoughts and behaviors: A systematic review. *Frontiers in Psychiatry*, 15:1291362, 2024.

[51] Nicholas B. Allen, Benjamin W. Nelson, David Brent, and Randy P. Auerbach. Short-term prediction of suicidal thoughts and behaviors in adolescents: Can recent developments in technology and computational science provide a breakthrough? *Journal of Affective Disorders*, 250:163–169, 2019.

[52] Edwin D. Boudreaux, Elke Rundensteiner, Feifan Liu, Bo Wang, Celine Larkin, and Emmanuel Agu. Applying machine learning approaches to suicide prediction using healthcare data: Overview and future directions. *Frontiers in Psychiatry*, 12:707916, 2021.

[53] Ewa K. Czyz, Cheryl A. King, and Inbal Nahum-Shani. Predicting short-term suicidal thoughts in adolescents using machine learning: Developing decision tools to identify daily level risk after hospitalization. *Psychological Medicine*, 53(12):6132–6142, 2023.

[54] Randy P. Auerbach, Richard Lan, Hanga Galfalvy, Kristine J. Alqueza, Jeffrey F. Cohn, Rebecca N. Crowley, Katherine Durham, Kaitlyn J. Joyce, Lisa E. Kahn, Rhea A. Kamath, Louis-Philippe Morency, Giovanna Porta, Anusha Srinivasan, Jennifer Zelazny, David A. Brent, and Nicholas B. Allen. Intensive longitudinal assessment of adolescents to predict suicidal thoughts and behaviors. *Journal of the American Academy of Child and Adolescent Psychiatry*, 62(9):1010–1020, 2023.

[55] Shaoxiong Ji, Guodong Long, Shirui Pan, Tianqing Zhu, Jing Jiang, and Shasha Wang. Detecting suicidal ideation with data protection in online communities. In *Database Systems for Advanced Applications*, volume 11448 of *Lecture Notes in Computer Science*, pages 225–229. Springer, 2019.

[56] Nikhileswar Komati. Suicide and depression detection. https://www.kaggle.com/datasets/nikhileswarkomati/suicide-watch, 2021. Accessed: 21 April 2025.

[57] Maleesha De Silva. Preprocessing steps for natural language processing (nlp): A beginner's guide. https://medium.com/@maleeshadesilva21/

preprocessing-steps-for-natural-language-processing-nlp-a-beginners-guide-d6d9
2023. Accessed: April 13, 2025.

[58] Kushankur Ghosh, Colin Bellinger, Roberto Corizzo, Paula Branco, Bartosz Krawczyk, and Nathalie Japkowicz. The class imbalance problem in deep learning. *Machine Learning*, 113(7):4845–4901, 2024.

[59] Eleni Alogogianni and Maria Virvou. Handling class imbalance and class overlap in machine learning applications for undeclared work prediction. *Electronics*, 12(4):913, 2023.

[60] Nazim Uddin Niaz, KM Nadim Shahariar, and Muhammed JA Patwary. Class imbalance problems in machine learning: A review of methods and future challenges. In *Proceedings of the 2nd International Conference on Computing Advancements*, pages 485–490, 2022.

Statement

I, **Shakib Al Hasan**, Student ID: **2021521460124**, hereby declare that the thesis entitled *"Developing Intelligence Systems for Detecting Suicidal Tendencies Using Transformer Models"* is the result of my independent research work, carried out under the supervision of **Dr. Dasha Hu**, Associate Professor, College of Software Engineering, Sichuan University.

To the best of my knowledge, this thesis does not contain any material previously published or written by any other person, nor does it contain any material that has been accepted for the award of any other degree or diploma at any academic institution, except where proper acknowledgment has been made in the text.

All sources of information, data, and scholarly contributions by others have been duly cited. Any collaborative work or assistance received during this research has been clearly stated and acknowledged within the thesis.

I further affirm that the research findings and results presented in this work were conducted during my undergraduate studies at Sichuan University and remain the intellectual property of Sichuan University.

Candidate's Signature: ________________________ **Date:** ______________

Name: Shakib Al Hasan

Student ID: 2021521460124

Program: Bachelor of Engineering in Software Engineering

College of Software Engineering, Sichuan University

Supervisor's Signature: ________________________ **Date:** ______________

Name: Dr. Dasha Hu

Associate Professor

College of Software Engineering, Sichuan University

Acknowledgement

First and foremost, I would like to express my deepest gratitude to my respected supervisor, **Dr. Dasha Hu**, Associate Professor at the College of Software Engineering, Sichuan University. His unwavering guidance, insightful feedback, and continuous encouragement have been instrumental throughout every stage of this thesis. Her expertise in the field, coupled with her patience and mentorship, has profoundly shaped the quality and direction of my research.

I am also sincerely thankful to the faculty members and staff of the College of Software Engineering at Sichuan University. Their academic support and the collaborative learning environment they fostered have greatly contributed to the successful completion of my undergraduate journey.

I extend my heartfelt appreciation to my family and friends for their constant motivation, emotional support, and understanding during the challenging phases of this academic endeavor. Their belief in me has been a source of strength and resilience.

Lastly, I would like to acknowledge all individuals—both directly and indirectly—who have contributed to this thesis. Their support, encouragement, and cooperation have been invaluable and truly appreciated.

May 2025 ***Shakib Al Hasan***

Bachelor of Engineering in Software Engineering

College of Software Engineering, Sichuan University